

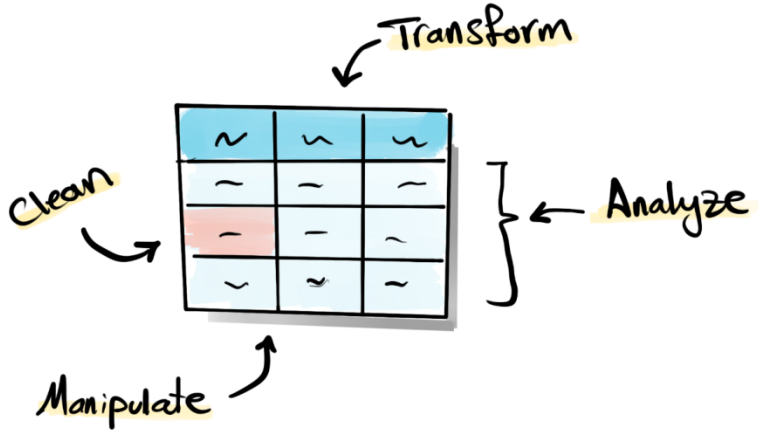
DATA WITH BARAA

SQL FUNCTIONS

Baraa Khatib Salkini
DATA WITH BARAA
SQL Course | SQL Functions



What are Functions?



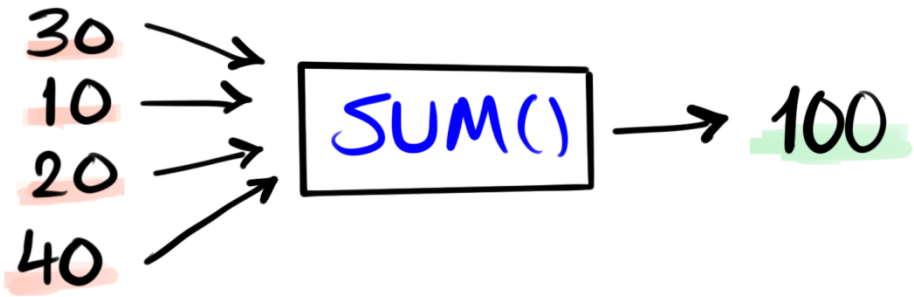
DATA WITH BARAA



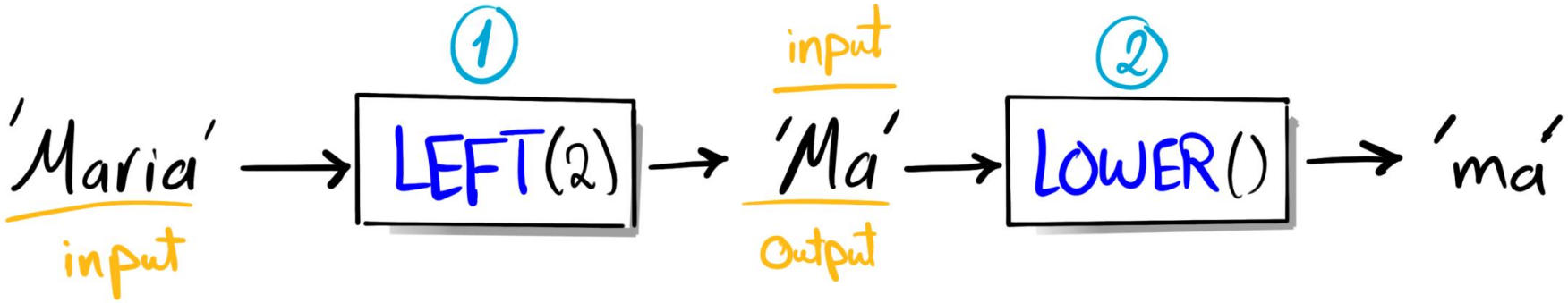
1. Single-Row Functions

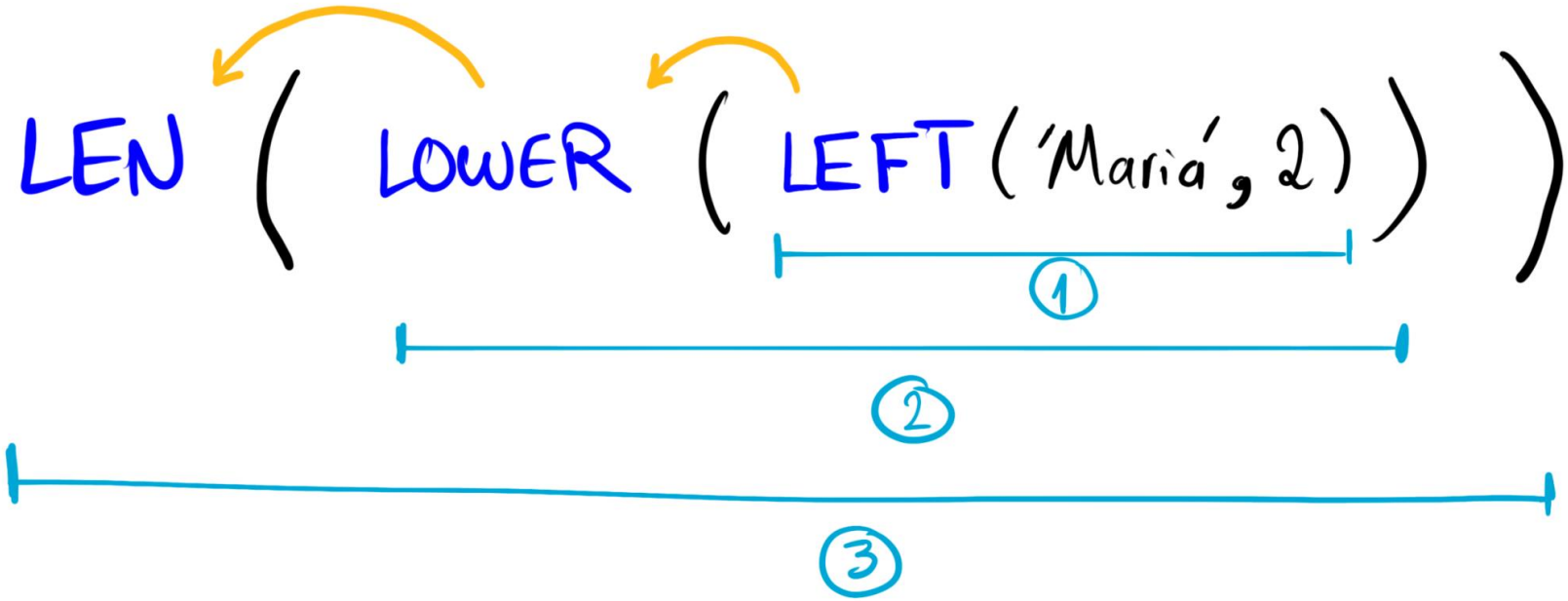


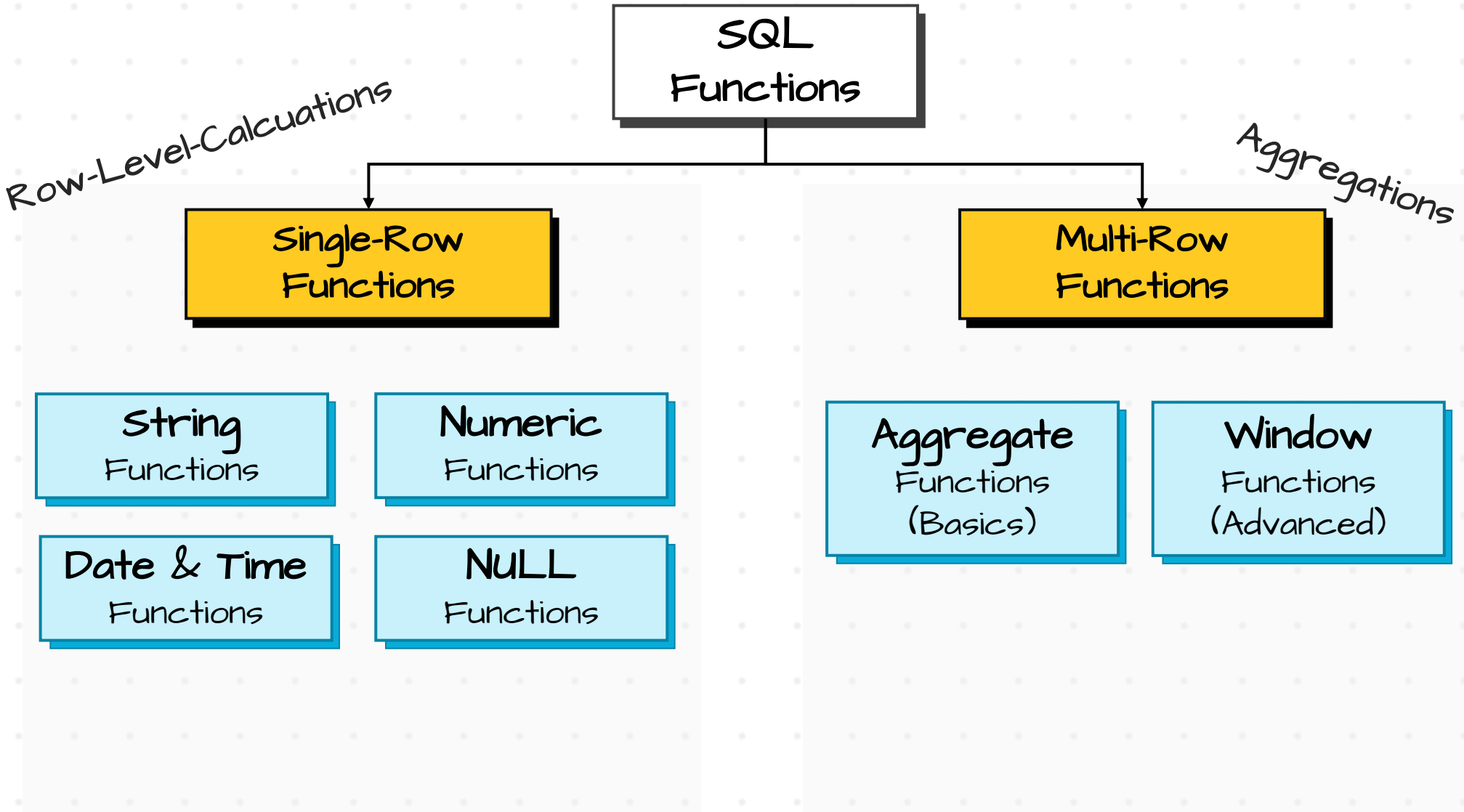
2. Multi-Row Functions

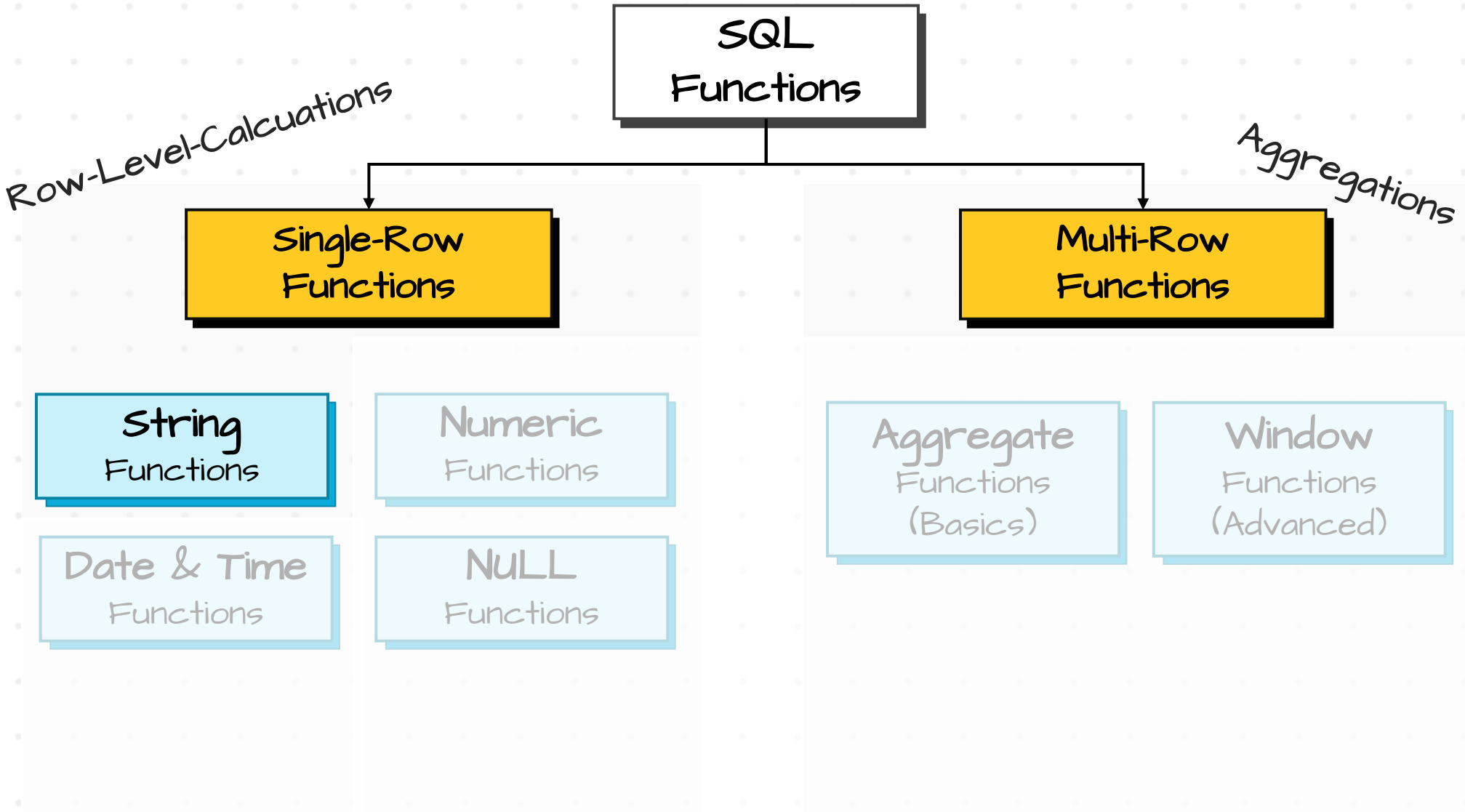


Nested Function






SQL Functions
Row-Level-Calcuations
Aggregations
Single-Row Functions
Multi-Row Functions
String Functions
Numeric Functions
Date & Time Functions
NULL Functions
Aggregate Functions (Basics)
Window Functions (Advanced)


SQL Functions
Row-Level-Calcuations
Aggregations
Single-Row Functions
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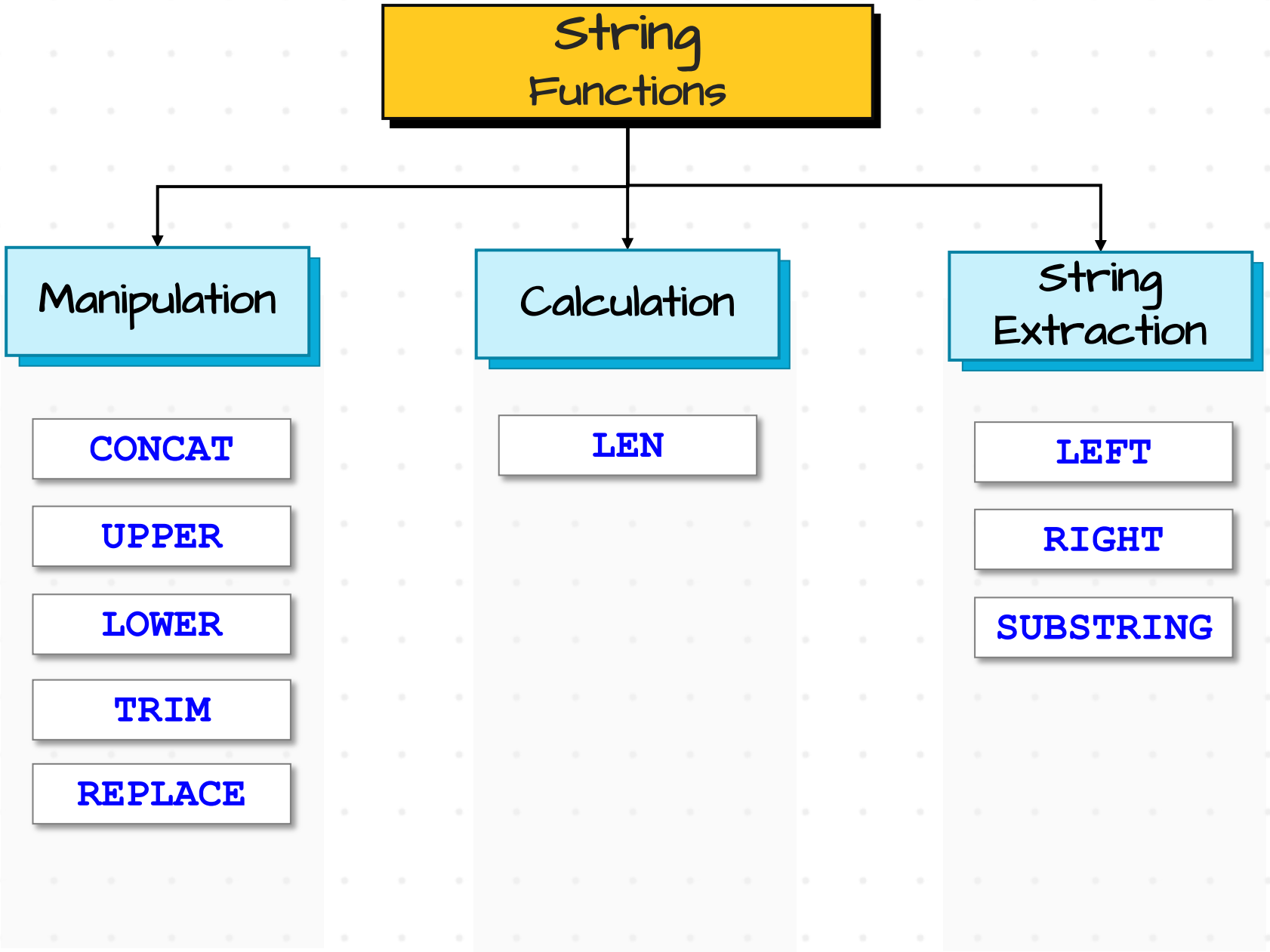


DATA WITH BARAA

STRING FUNCTIONS

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DATA WITH BARAA
SQL Course | String Functions





DATA WITH BARAA

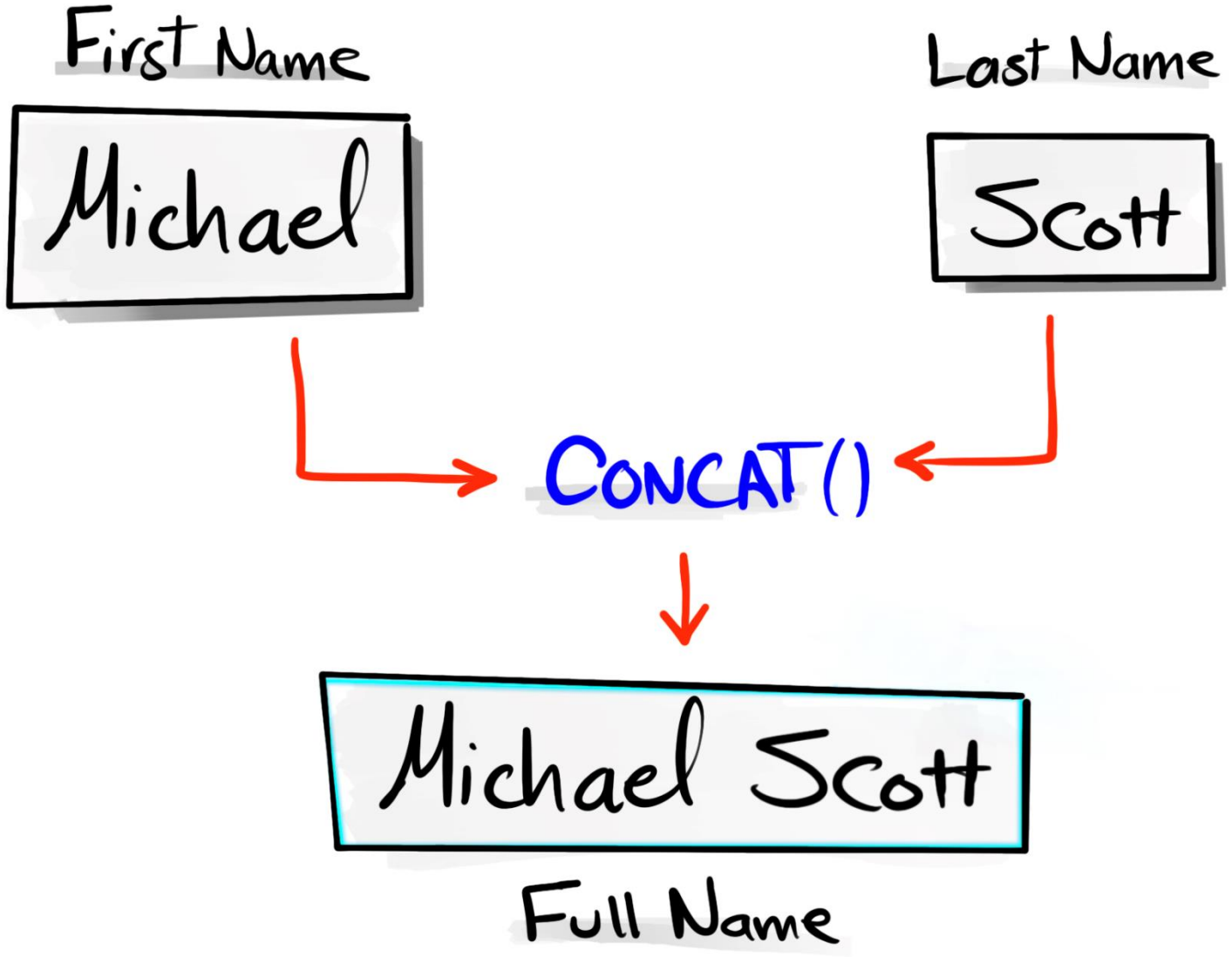
String Functions

CONCAT	Combines multiple strings into one
UPPER	Converts all characters to uppercase
LOWER	Converts all characters to lowercase
TRIM	Removes Leading and Trailing spaces
REPLACE	Replaces specific character with a new character
LEN	Counts how many characters
LEFT	Extracts specific Number of Characters from the start
RIGHT	Extracts specific Number of Characters from the End
Substring	Extracts a part of string at a specified position

DATA WITH BARAA

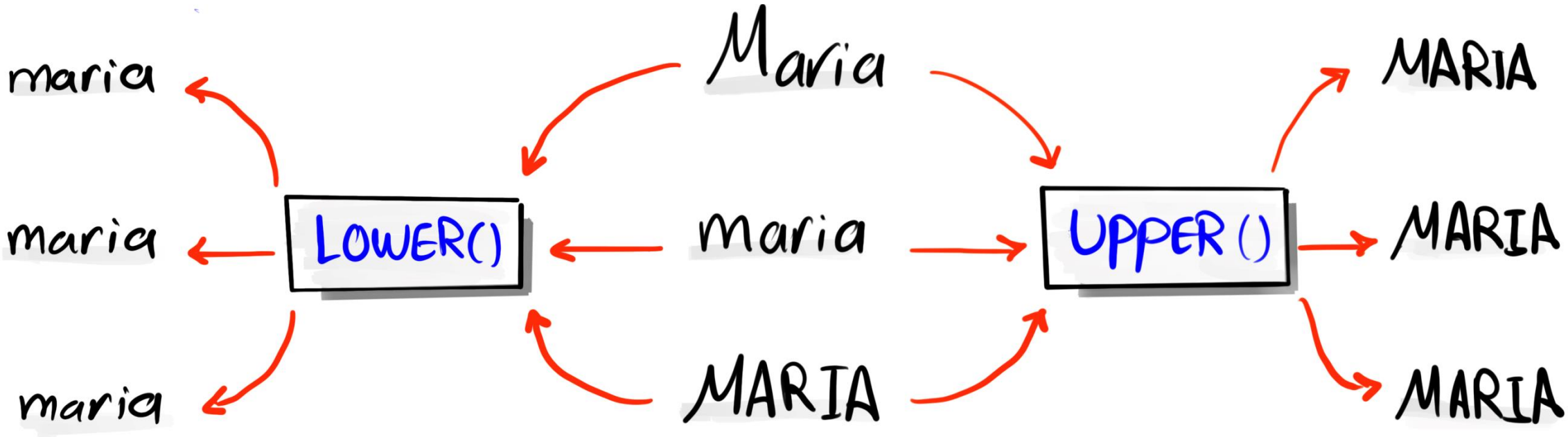
CONCAT

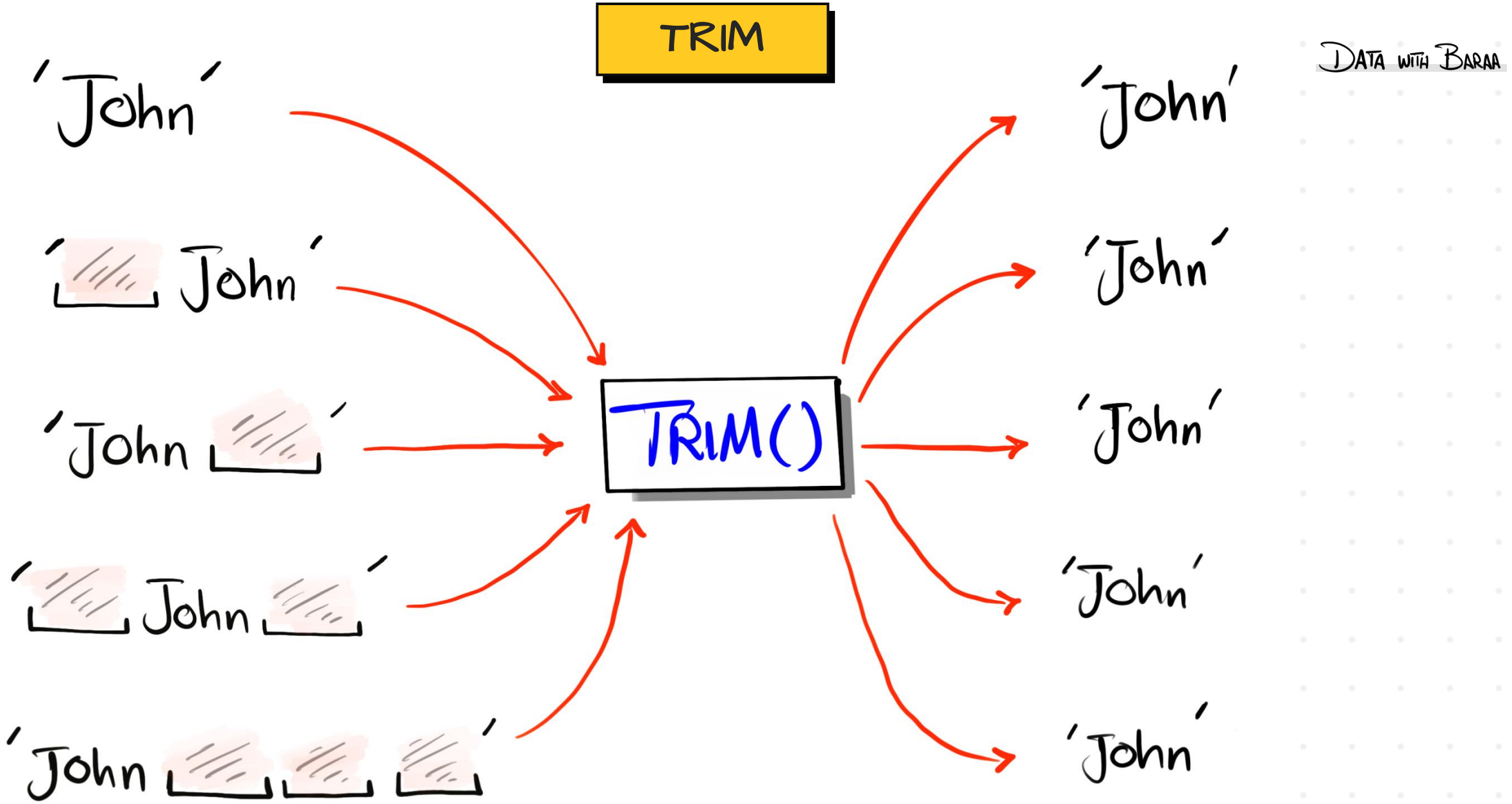
DATA WITH BARAA



LOWER & UPPER

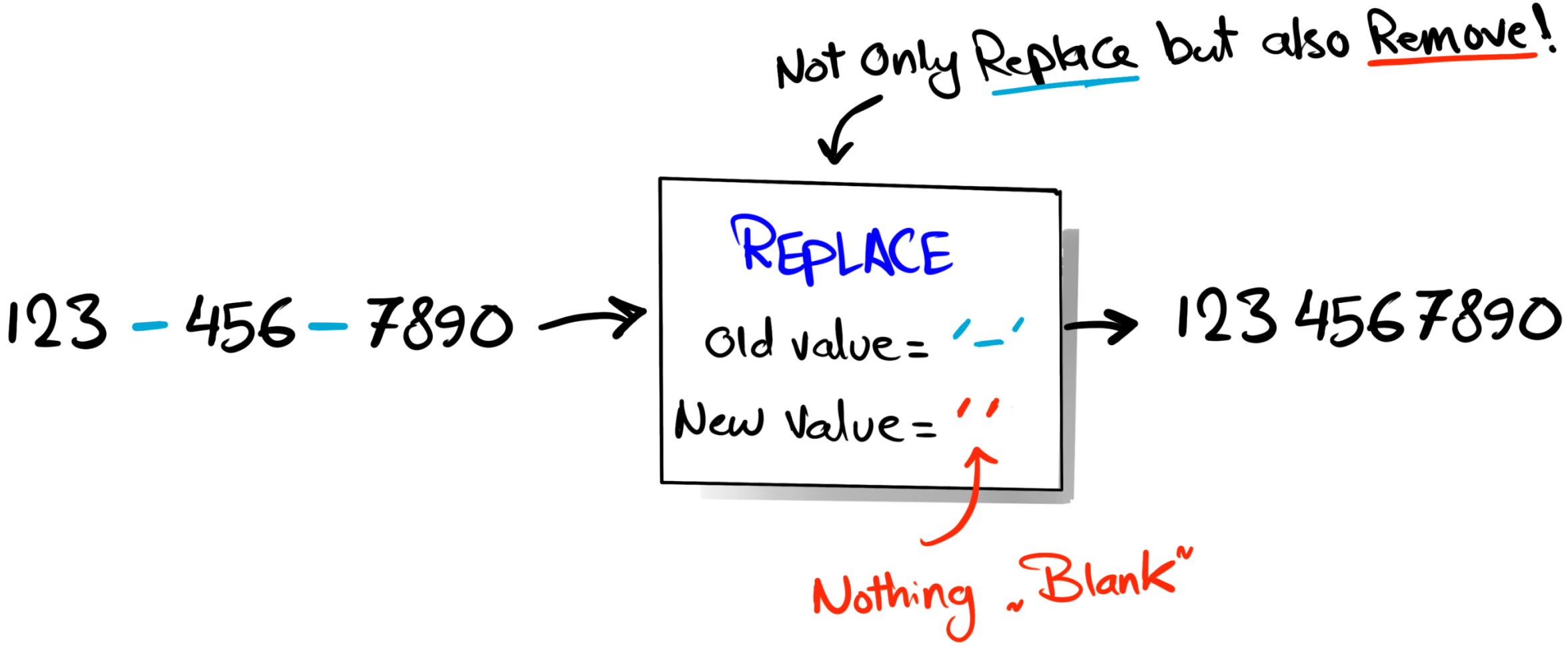
DATA WITH BARAA




TRIM
DATA WITH BARAA
'John'
' John'
'John '
' John '
' John '
TRIM()
'John'
'John'
'John'
'John'
'John'

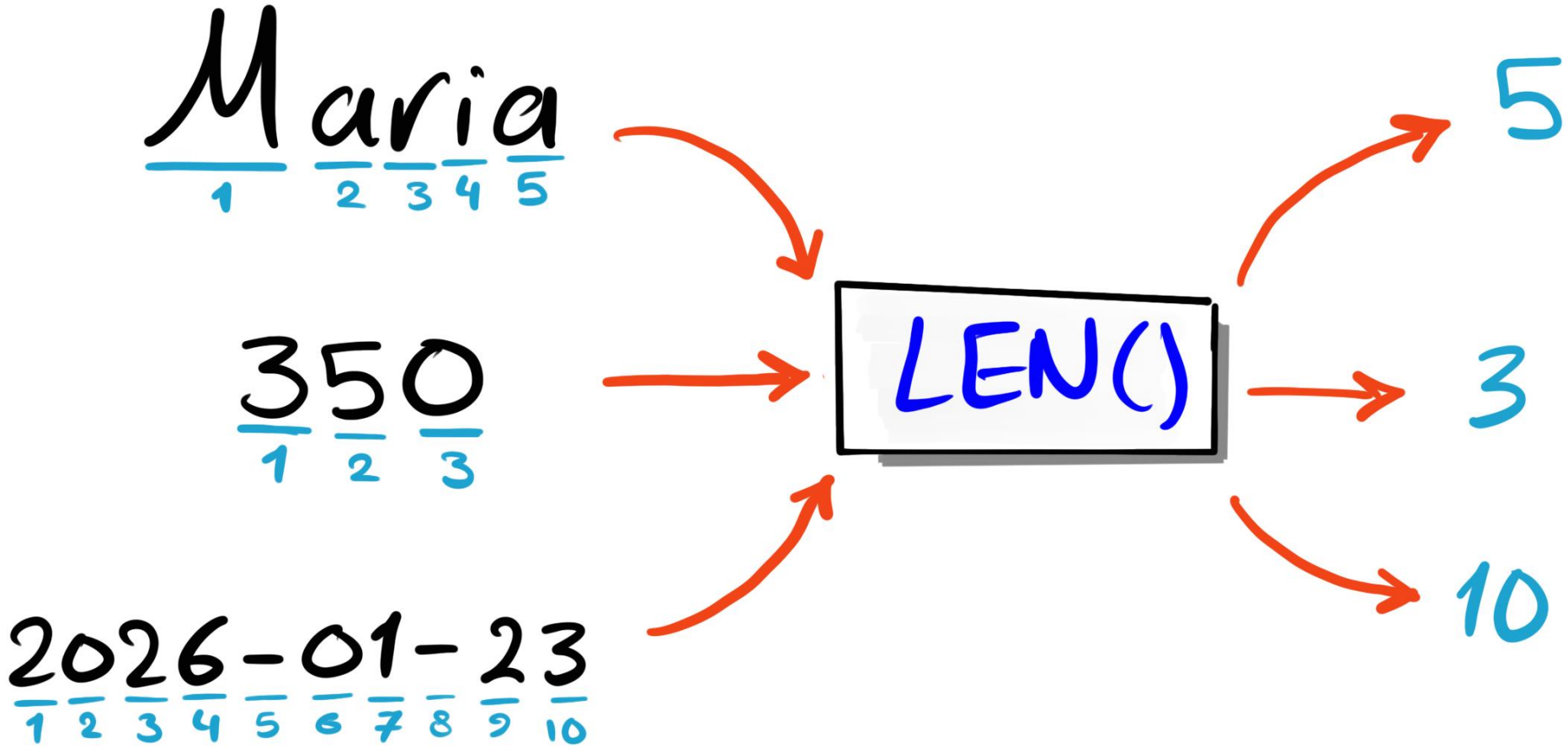
REPLACE

DATA WITH BARAA



LEN

DATA WITH BARAA



LEFT & RIGHT

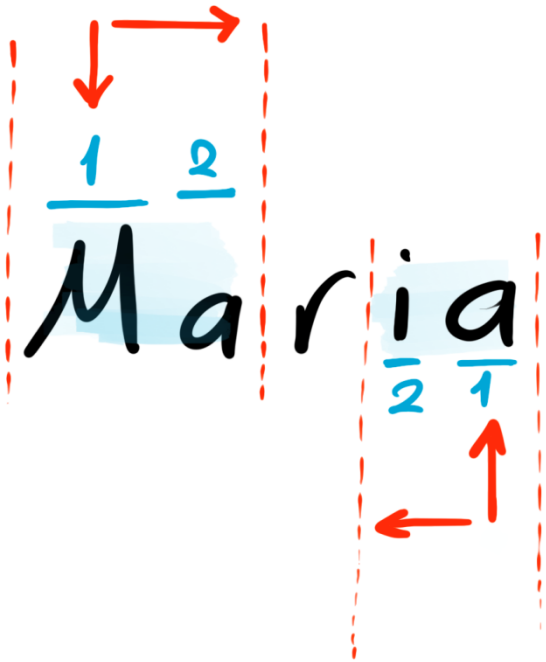
LEFT (Value , Nr of Characters)

RIGHT (Value , Nr of Characters)

Extract
First 2 Characters

LEFT = 2

Ma



Extract
Last 2 Characters

RIGHT = 2

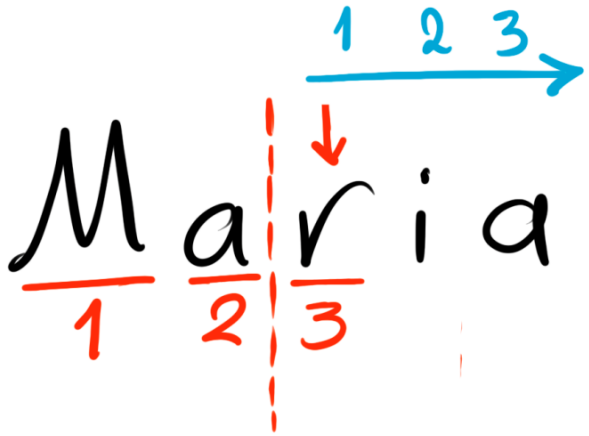
ia

SUBSTRING (Value , Start , Length)

DATA WITH BARAA

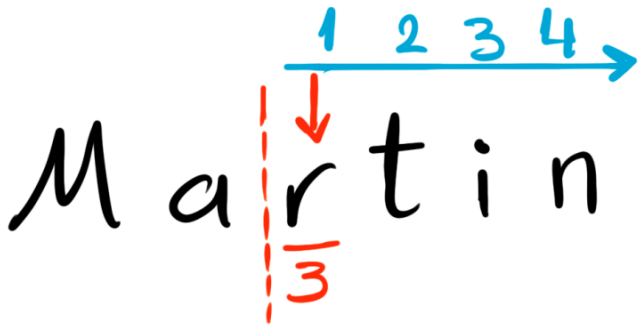
After the 2nd Character extract All Characters

SUBSTRING



Start
3

Length
← Dynamic
LEN()
5
6
10



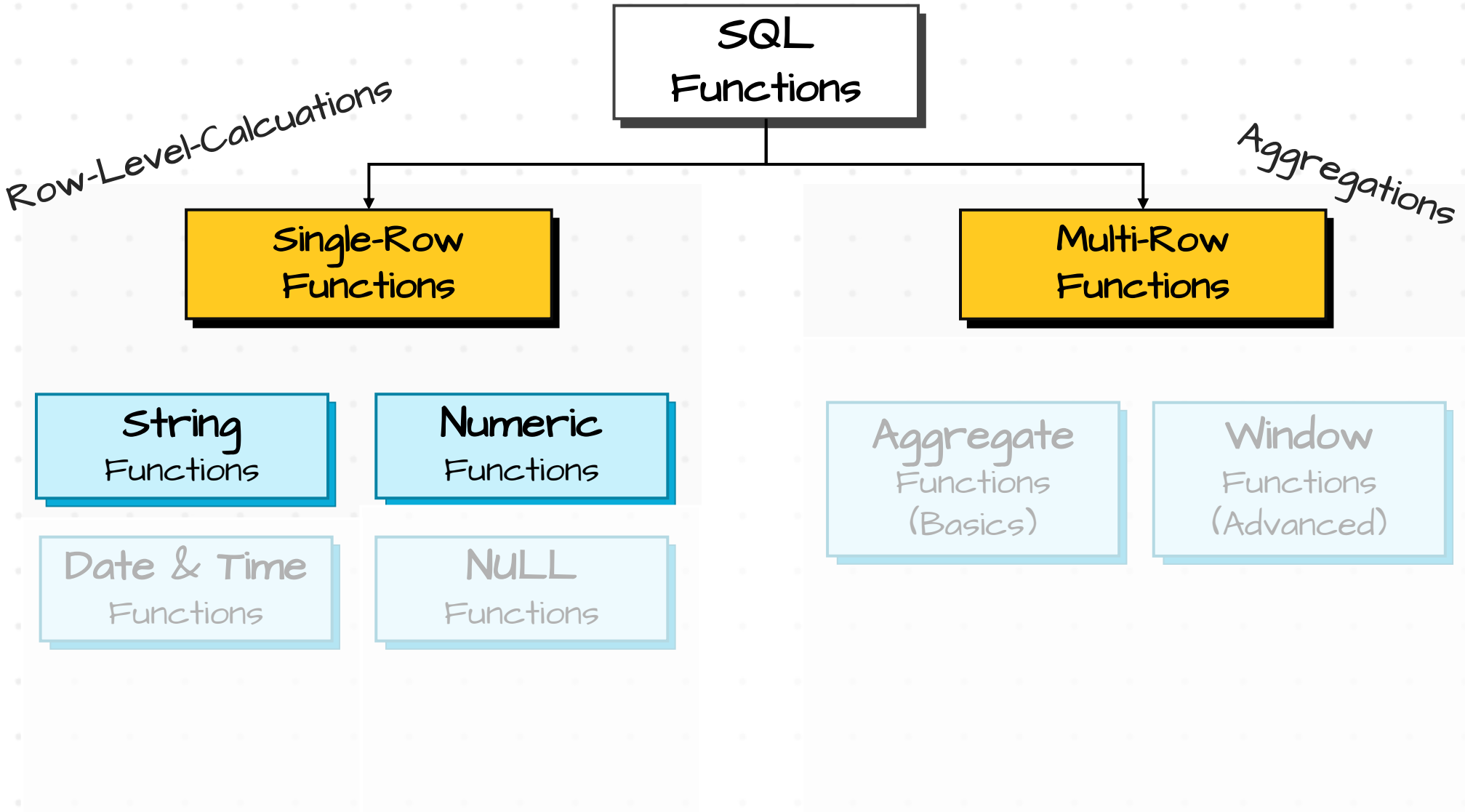


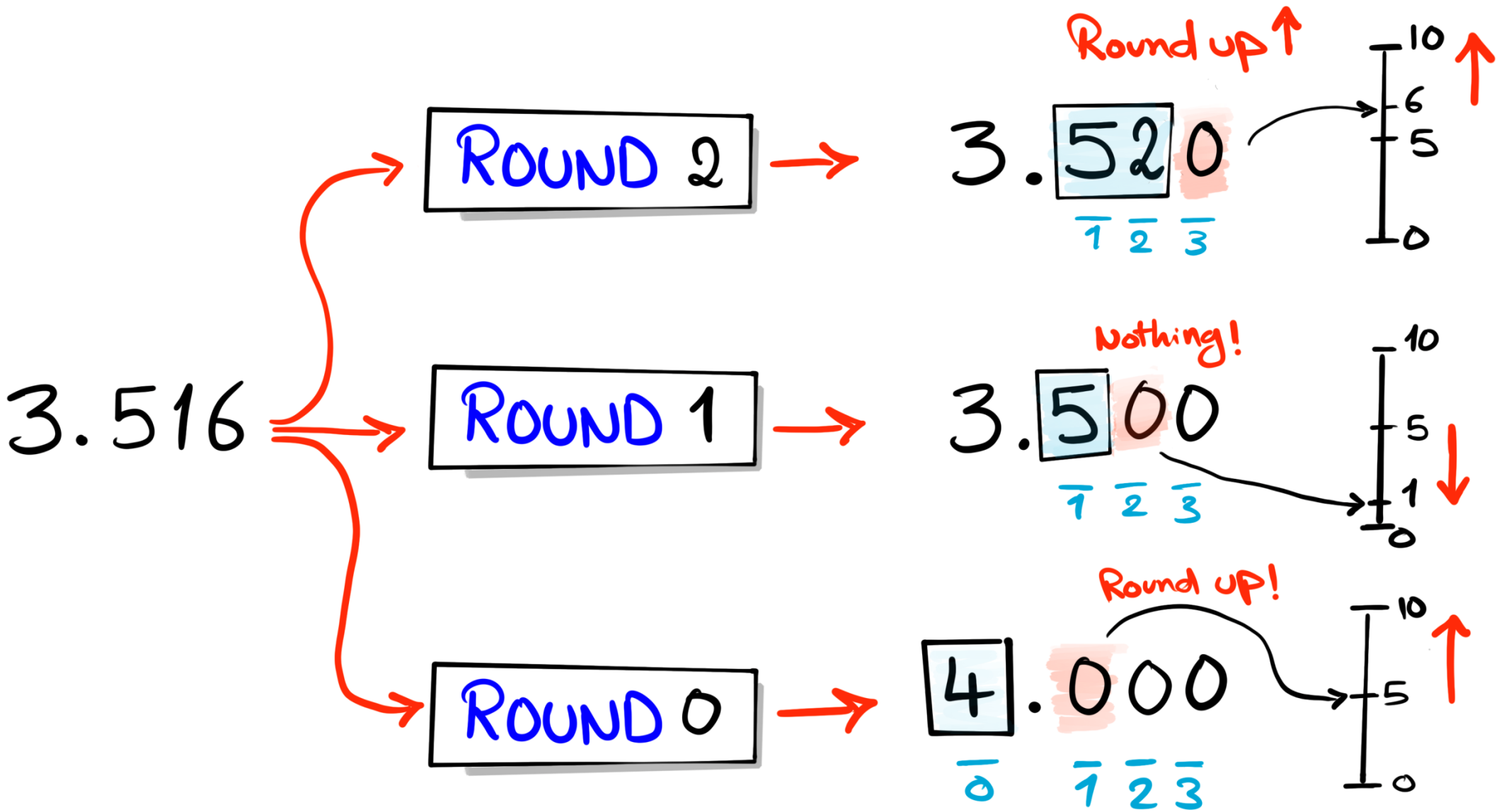
DATA WITH BARAA

NUMERIC FUNCTIONS

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DATA WITH BARAA
SQL Course | Number Functions




SQL Functions
Row-Level-Calcuations
Aggregations
Single-Row Functions
Multi-Row Functions
String Functions
Numeric Functions
Date & Time Functions
NULL Functions
Aggregate Functions (Basics)
Window Functions (Advanced)


3.516
ROUND 2
ROUND 1
ROUND 0
Round up
3.520
1 2 3
10
6
5
0
Nothing!
3.500
1 2 3
10
5
1
0
Round up!
4.000
0 1 2 3
10
5
0



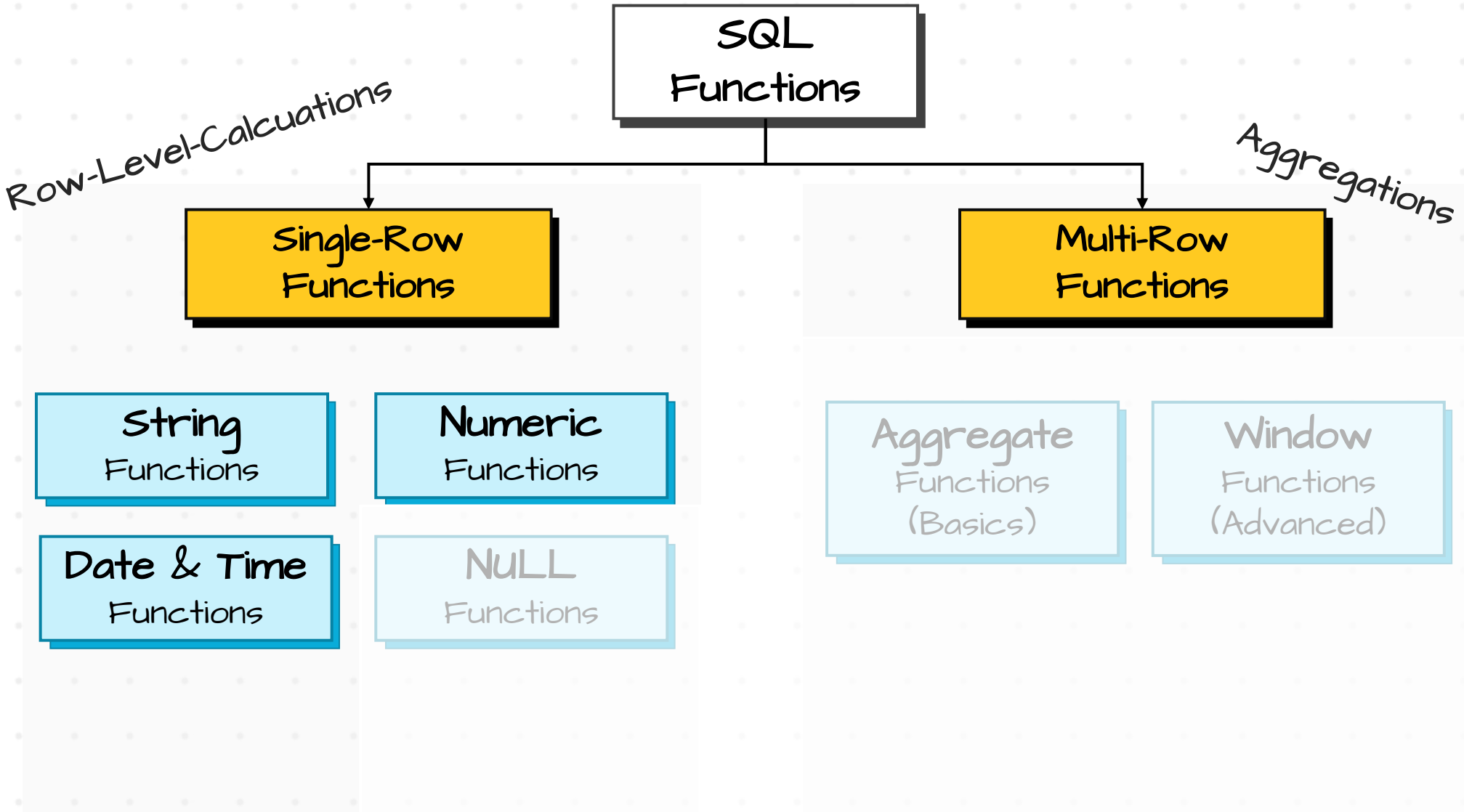
DATA WITH BARAA

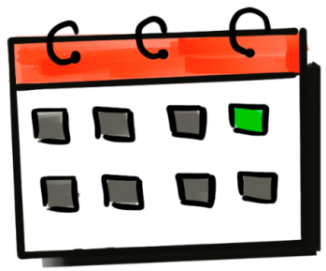
DATE & TIME

FUNCTIONS

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DATA WITH BARAA
SQL Course | Date & Time Funcitons




SQL Functions
Row-Level-Calcuations
Aggregations
Single-Row Functions
Multi-Row Functions
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Aggregate Functions (Basics)
Window Functions (Advanced)



Date

2025 - 08 - 20

Year | Month | Day

Time

18 : 55 : 45

Hours | Minutes | Seconds





Timestamp

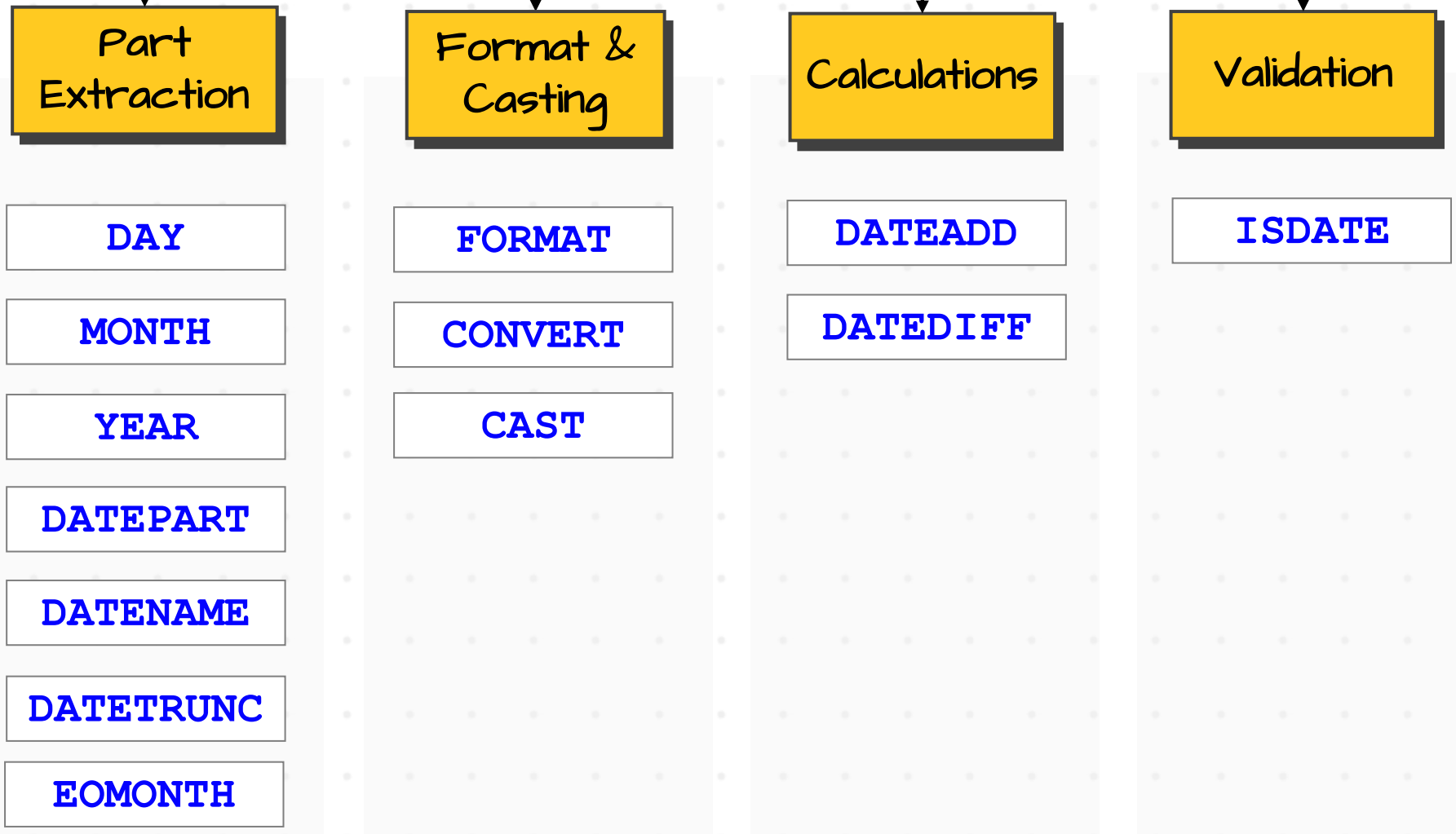
(Oracle, Postgres, MySQL)

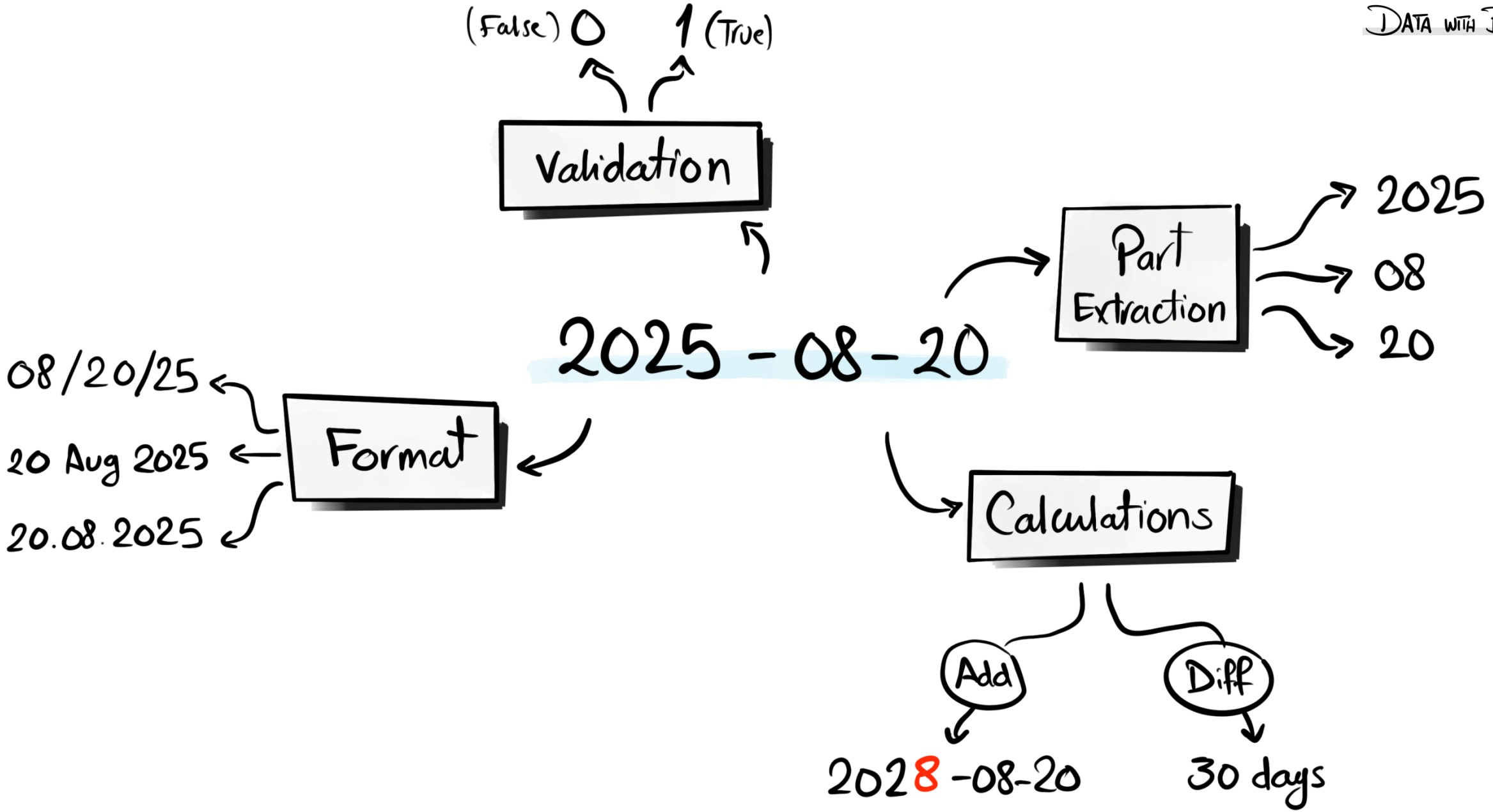
Datetime2

(SQL Server)

Date & Time Functions

DATA WITH BARAA




(False) 0
1 (True)
Validation
2025-08-20
Part Extraction
2025
08
20
08/20/25
20 Aug 2025
20.08.2025
Format
Calculations
Add
Diff
2028-08-20
30 days



DATA WITH BARAA

PARTS EXTRACTION

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DATA WITH BARAA
SQL Course | Date & Time Funcitons

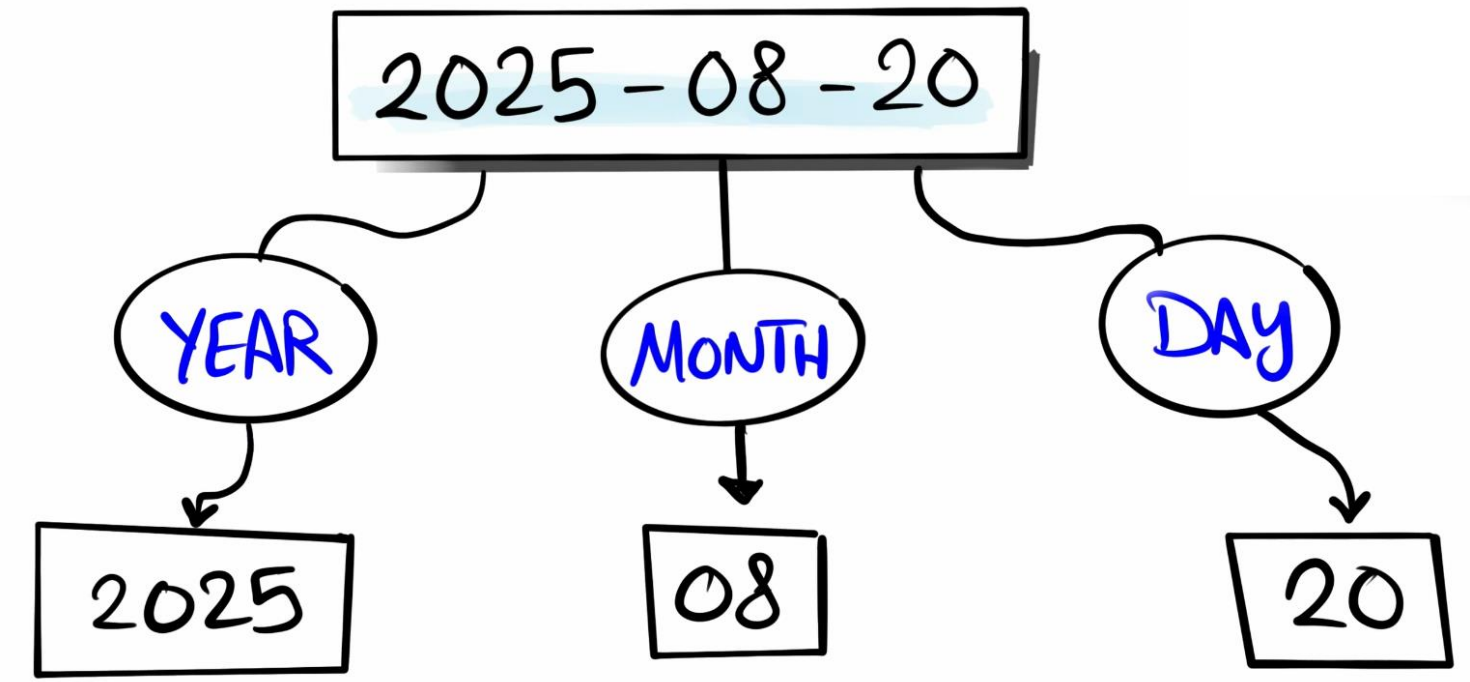


Date & Time Functions

DATA WITH BARAA

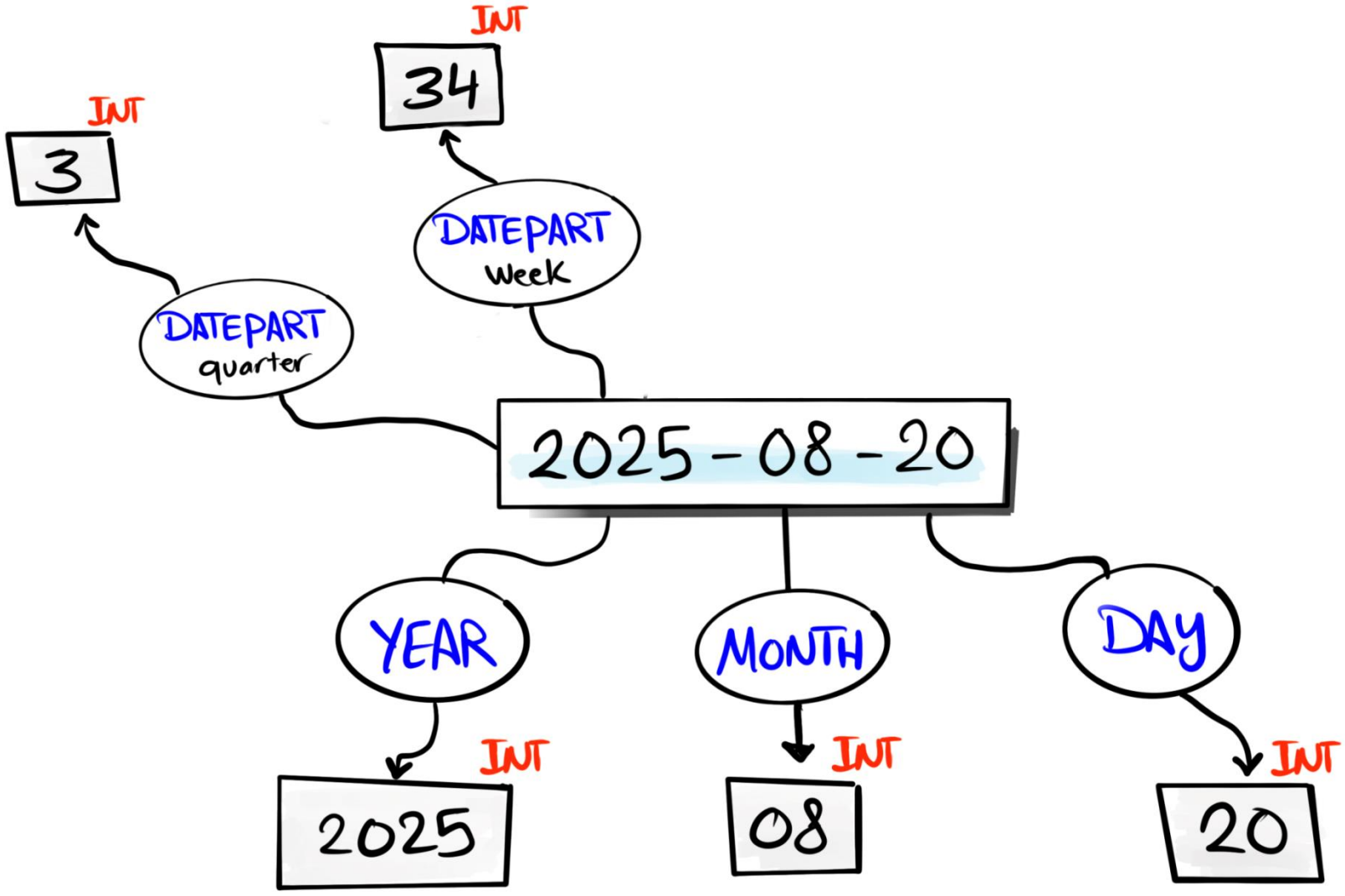
- **Part Extraction**
  - DAY
  - MONTH
  - YEAR
  - DATEPART
  - DATENAME
  - DATETRUNC
  - EOMONTH
- Format & Casting
  - FORMAT
  - CONVERT
  - CAST
- Calculations
  - DATEADD
  - DATEDIFF
- Validation
  - ISDATE

Quick Functions
YEAR, MONTH, DAY



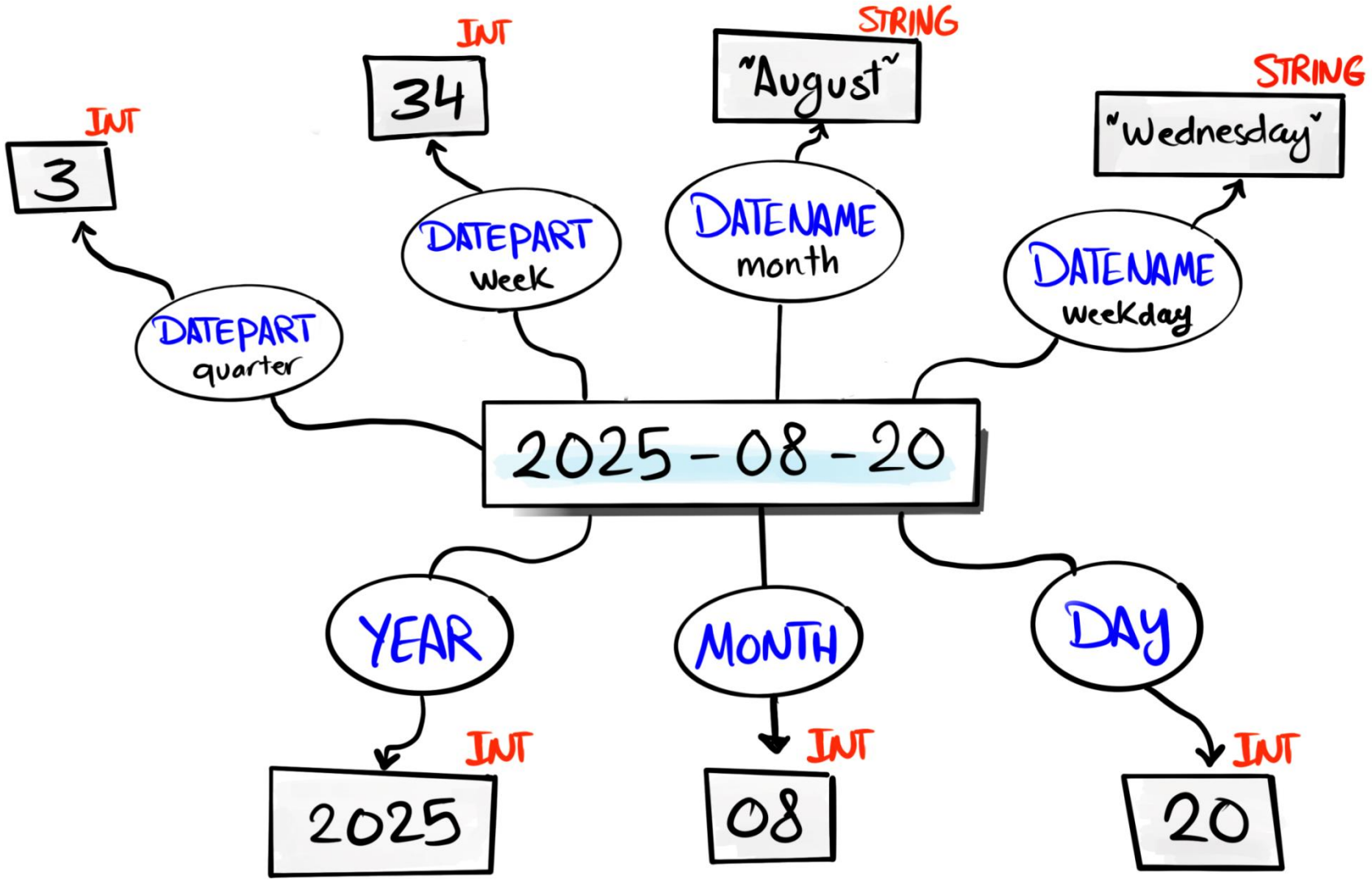
DATEPART

DATA WITH BARAA



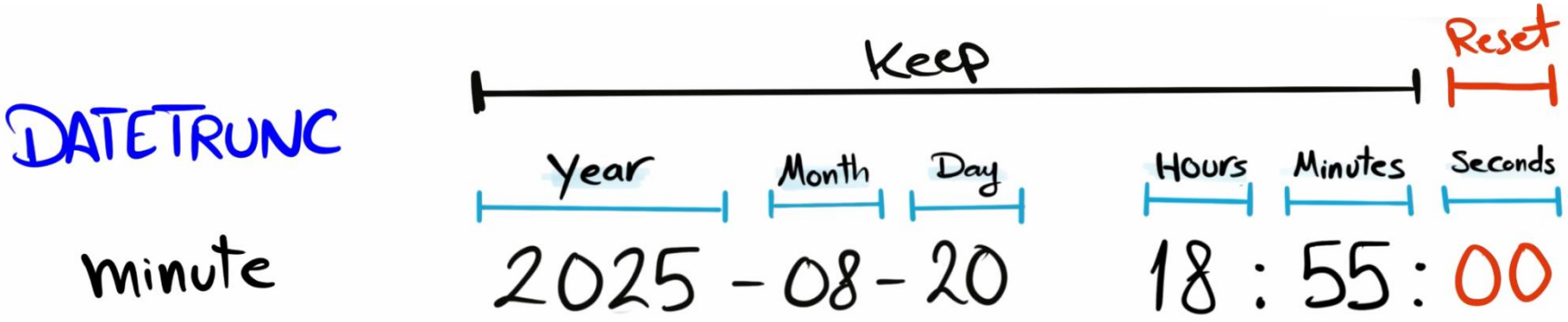
DATENAME

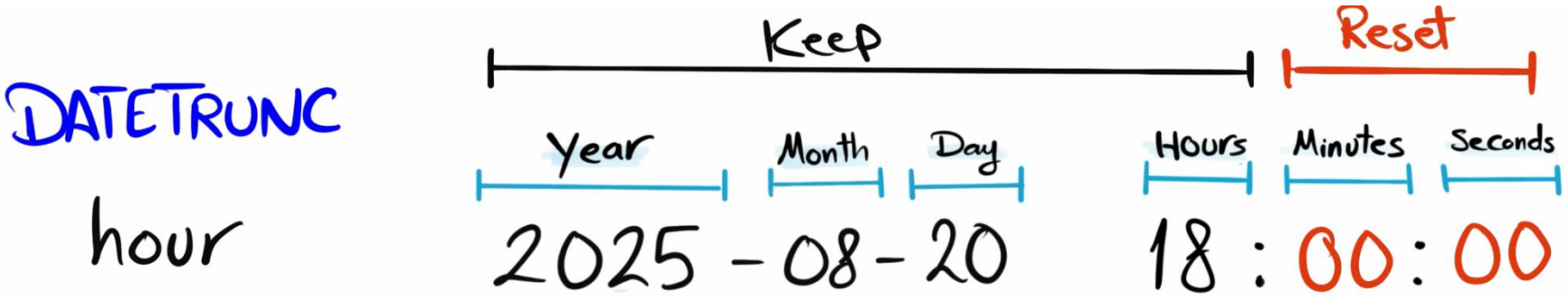
DATA WITH BARAA



DATETRUNC

DATA WITH BARAA





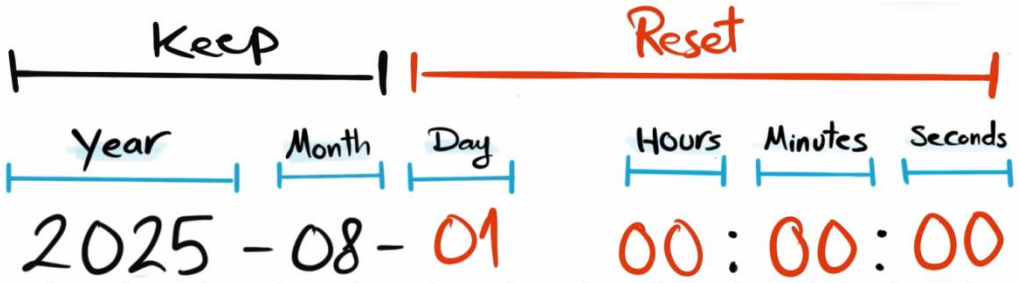
DATETRUNC

DATA WITH BARAA

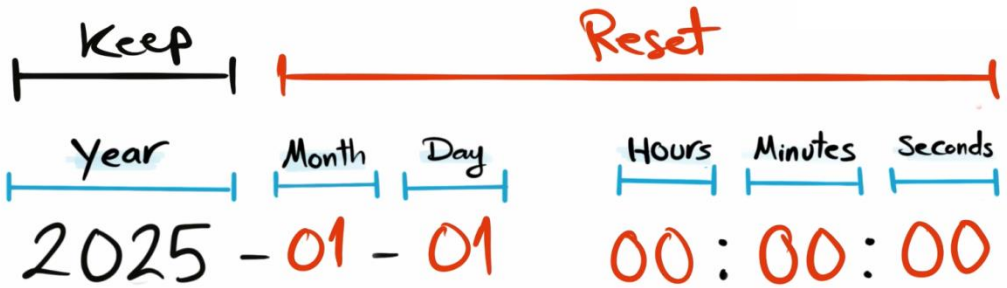
DATETRUNC day



DATETRUNC month



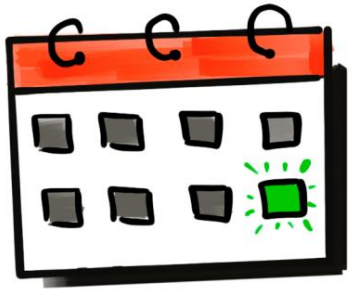
DATETRUNC year



Datepart resets to 01

Time part resets to 00

EOMONTH



EOMONTH()

Year | Month | Day

2025 - 08 - 31

2025 - 02 - 28

2025 - 03 - 31

} DATE

PART EXTRACTION
Syntax


```
DAY(date)
```

```
MONTH(date)
```

```
YEAR(date)
```

```
EOMONTH(date)
```

```
DATEPART(part, date)
```

```
DATENAME(part, date)
```

```
DATETRUNC(part, date)
```

DATA TYPES

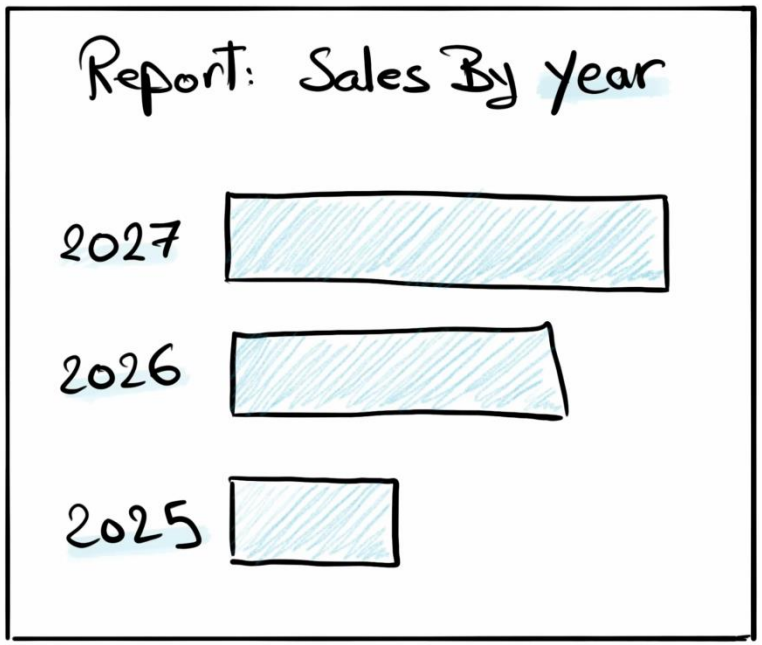
DATA WITH BARAA

	DATA TYPE
DAY MONTH YEAR DATEPART	INT
DATENAME	STRING
DATETRUNC	DATETIME
EOMONTH	DATE

DATE PARTS

DATA WITH BARAA

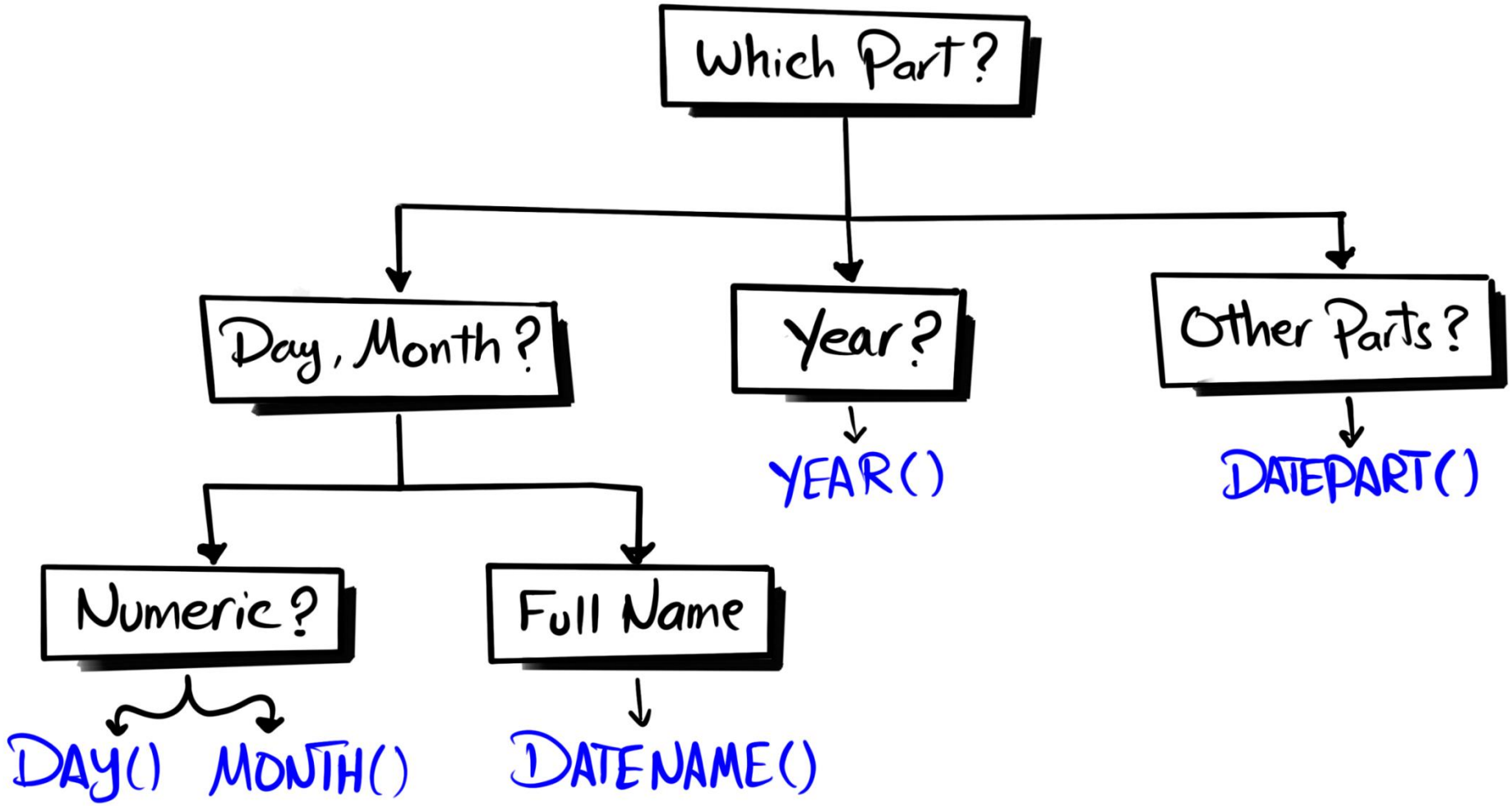
Date parts help to aggregate data at different levels of granularity (year, month, day, etc.).







How to Choose the Right Function?



2025-08-20
09:38:54.840

Date Parts

DATA WITH BARAA

DATEPART

DATENAME

DATETUNC

		INT	String	Datetime2
Part	Abbre.	DATEPART	DATENMAME	DATETRUNC
year	yy, yyyy	2025	2025	2025-01-01 00:00:00
quarter	qq,q	3	3	2025-07-01 00:00:00
month	mm,m	8	August	2025-08-01 00:00:00
dayofyear	dy,y	232	232	2025-08-20 00:00:00
day	dd, d	20	20	2025-08-20 00:00:00
weekday	dw	4	Wednesday	Not supported
week	wk,ww	34	34	2025-08-17 00:00:00
iso_week	ns	34	34	2025-08-18 00:00:00
hour	hh	9	9	2025-08-20 09:00:00
minute	mi,n	45	45	2025-08-20 09:45:00
second	ss,s	21	21	2025-08-20 09:45:21
millisecond	ms	0	0	2025-08-20 09:45:21
microsecond	msc	0	0	2025-08-20 09:45:21
nanosecond	ns	0	0	Not supported
iso_week	isowk, isoww	0	+00:00	Not supported



DATA WITH BARAA

DATE FORMATS

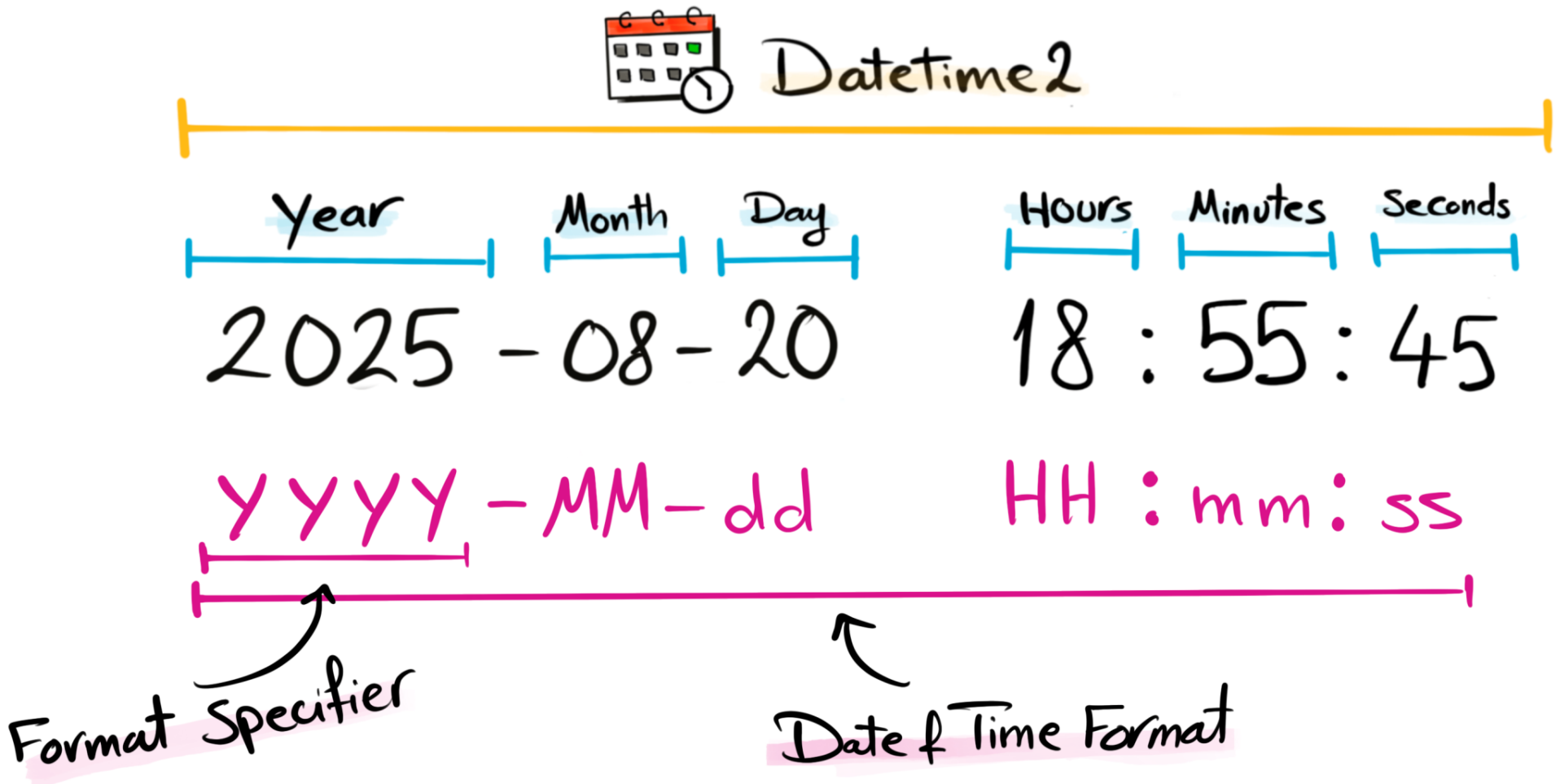
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DATA WITH BARAA
SQL Course | Date & Time Funcitons



Date & Time Functions

DATA WITH BARAA

- **Part Extraction**
  - DAY
  - MONTH
  - YEAR
  - DATEPART
  - DATENAME
  - DATETRUNC
  - EOMONTH
- **Format & Casting**
  - FORMAT
  - CONVERT
  - CAST
- **Calculations**
  - DATEADD
  - DATEDIFF
- **Validation**
  - ISDATE


Datetime2
Year
Month
Day
Hours
Minutes
Seconds
2025-08-20
18:55:45
YYYY-MM-dd
HH:mm:ss
Format Specifier
Date & Time Format

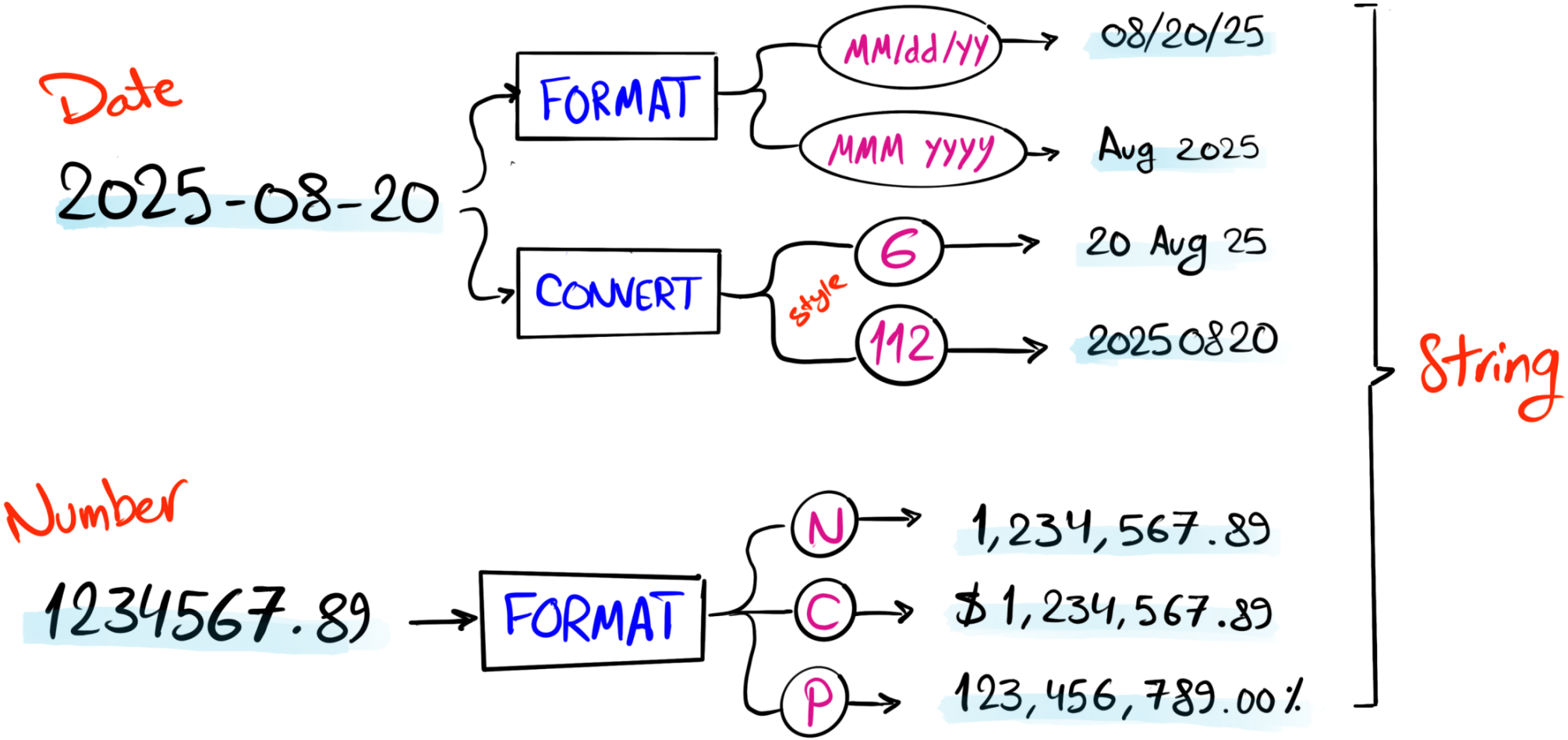


2025-08-20
YYYY-MM-dd
} International Standard (Iso 8601)

08-20-2025
MM-dd-YYYY
} USA Standard

20-08-2025
dd-MM-YYYY
} European Standard

FORMATING


Date
2025-08-20
FORMAT
MM/dd/yy
08/20/25
MMM yyyy
Aug 2025
CONVERT
style
6
20 Aug 25
112
20250820
String
Number
1234567.89
FORMAT
N
1,234,567.89
C
$1,234,567.89
P
123,456,789.00%

CAST()
CONVERT()

CASTING

"change Data Types"

String '123' → 123 Number

Date 2025-08-20 → '2025-08-20' String

String '2025-08-20' → 2025-08-20 Date

FORMAT Syntax

DATA WITH BARAA

Syntax

```
FORMAT(value, format [,culture])
```

Optional

Examples

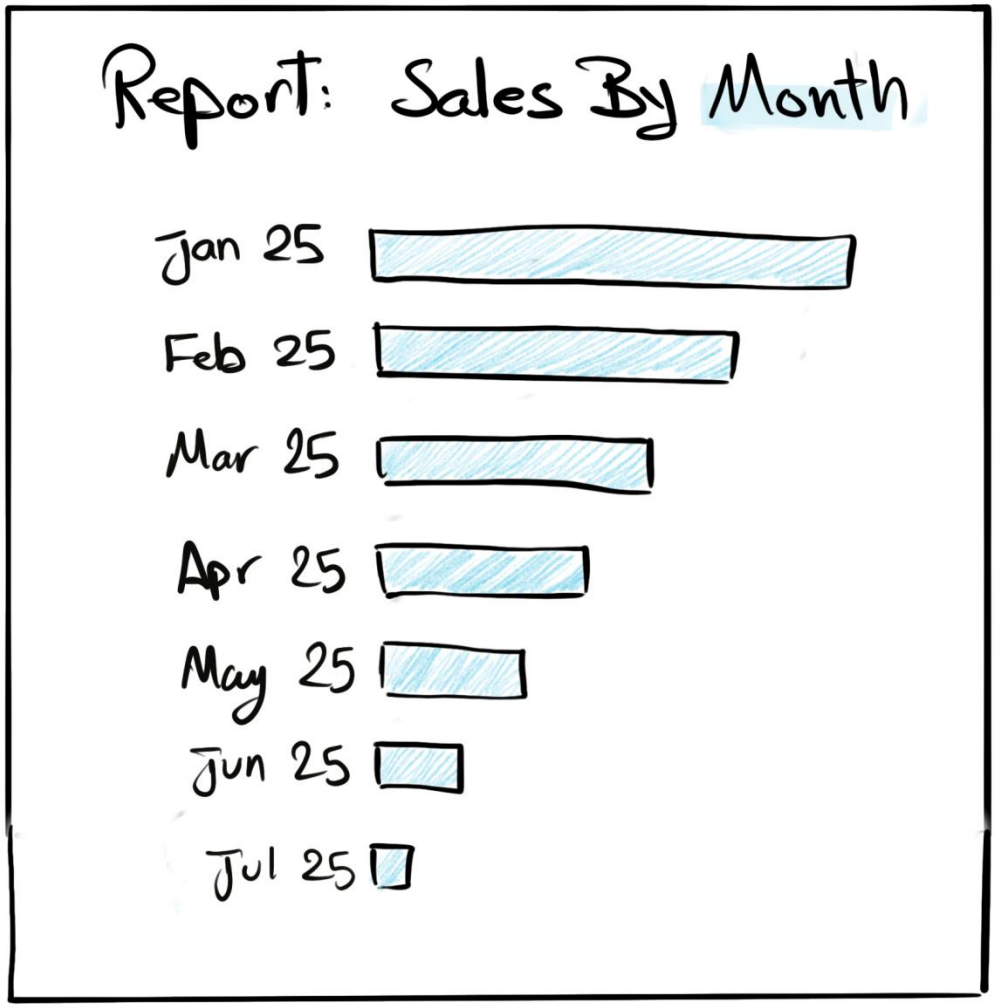
```
FORMAT(OrderDate,'dd/MM/yyyy')
```

```
FORMAT(OrderDate,'dd/MM/yyyy','ja-JP')
```

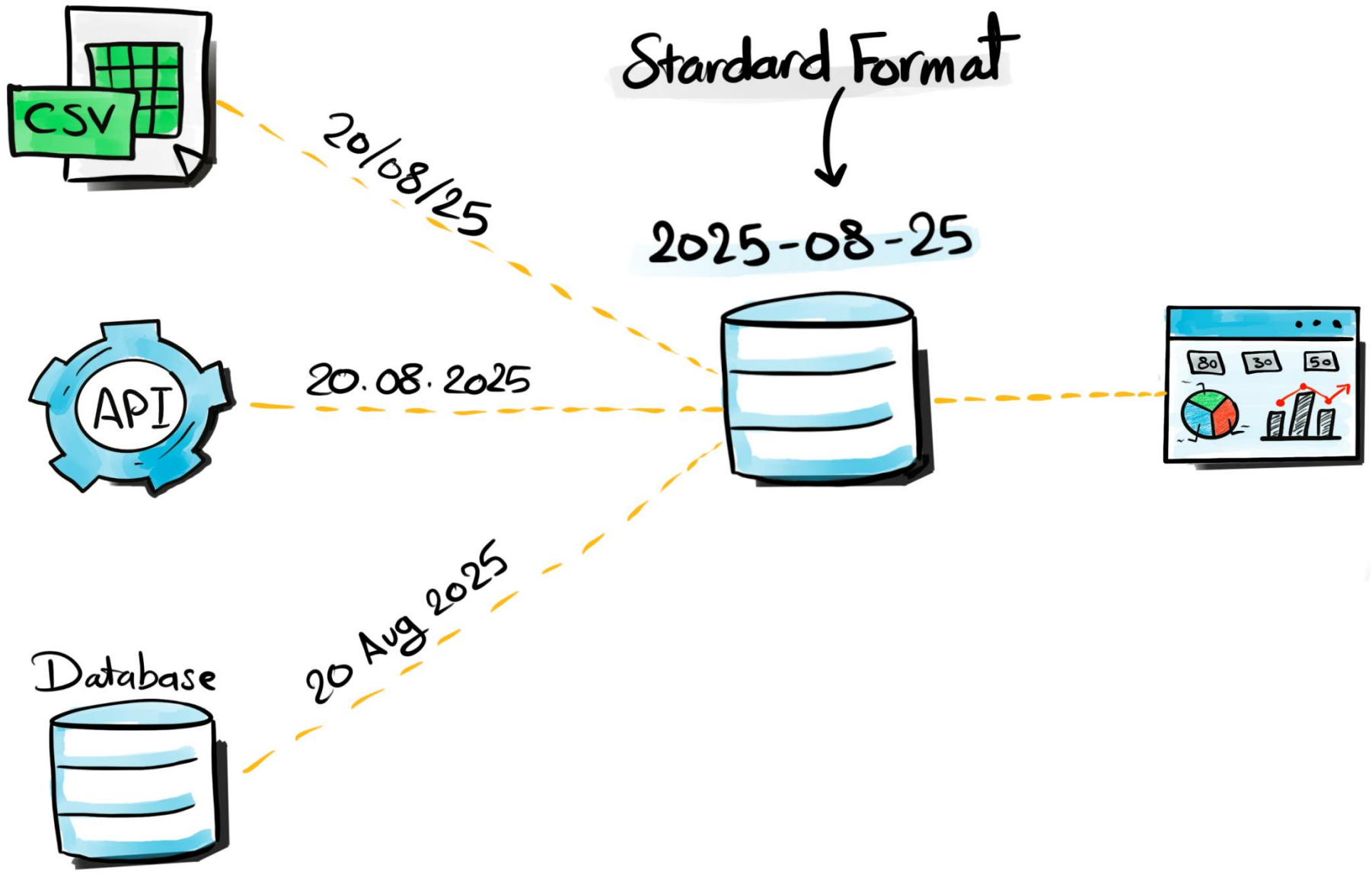
```
FORMAT(1234.56, 'D' ,'fr-FR')
```

Default Cultuer = 'en-US'

FORMAT
Use Case



FORMAT
Use Case



2025-08-20
18:55:45

Date & Time Format Speicifiers

DATA WITH BARAA

FORMAT

Format	Description	Result
D	Full day name	
d	Day of the month	8/20/2025
dd	Day of the month (two-digit)	20
ddd	Abbreviated day name	Wed
dddd	Full day name	Wednesday
M	Month number	44044
MM	Month number (two-digit)	8
MMM	Abbreviated month name	Aug
MMMM	Full month name	August
yy	Year (two-digit)	25
yyyy	Year (four-digit)	2025
hh	Hour (12-hour format, two-digit)	06
HH	Hour (24-hour format, two-digit)	18
m	Minutes	August 20
mm	Minutes (two-digit)	55
s	Seconds	2025-08-20T18:55:45
ss	Seconds (two-digit)	45
f	Fractional seconds (one digit)	Wednesday, August 20, 2025 6:55 PM
ff	Fractional seconds (two digits)	00
fff	Fractional seconds (three digits)	000
tt	AM/PM designator	PM

2025-08-20
18:55:45

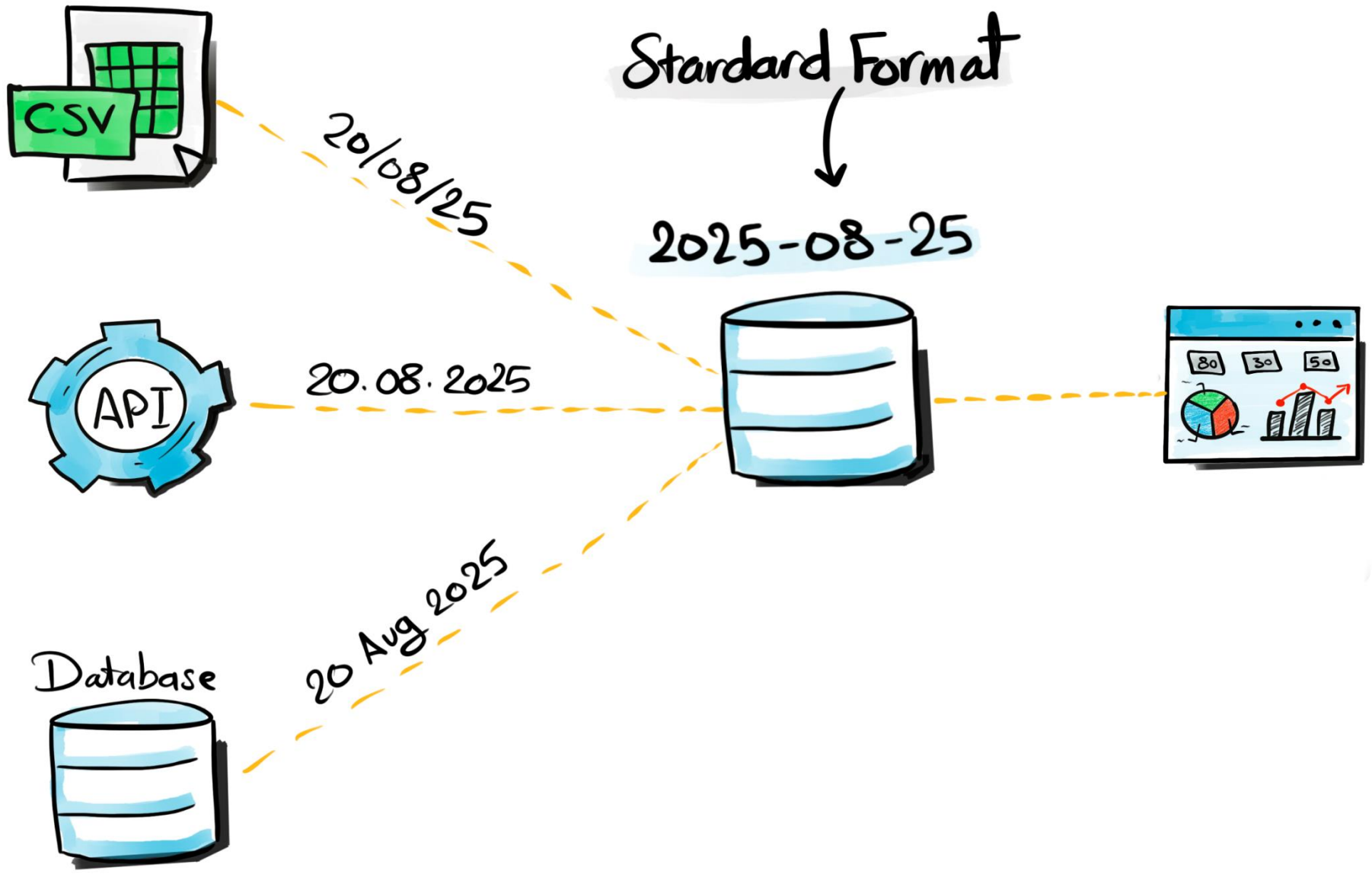
Number Format Speicifiers

DATA WITH BARAA

FORMAT

Format	Description	Query	Result
N	Numeric default	SELECT FORMAT(1234.56, 'N')	1,234.56
P	Percentage	SELECT FORMAT(1234.56, 'P')	123,456.00 %
C	Currency	SELECT FORMAT(1234.56, 'C')	$1,234.56
E	Scientific notation	SELECT FORMAT(1234.56, 'E')	1,23E+09
F	Fixed-point	SELECT FORMAT(1234.56, 'F')	1234.56
N0	Numeric no decimals	SELECT FORMAT(1234.56, 'N0')	1,235
N1	Numeric one decimal	SELECT FORMAT(1234.56, 'N1')	1,234.6
N2	Numeric two decimals	SELECT FORMAT(1234.56, 'N2')	1,234.56
N , de_DE	Numeric (German)	SELECT FORMAT(1234.56, 'N', 'de-DE')	1.234,56
N, en_US	Numeric (US)	SELECT FORMAT(1234.56, 'N', 'en-US')	1,234.56

FORMAT
Use Case



CONVERT

Syntax

```
CONVERT(data_type, value [,style])
```

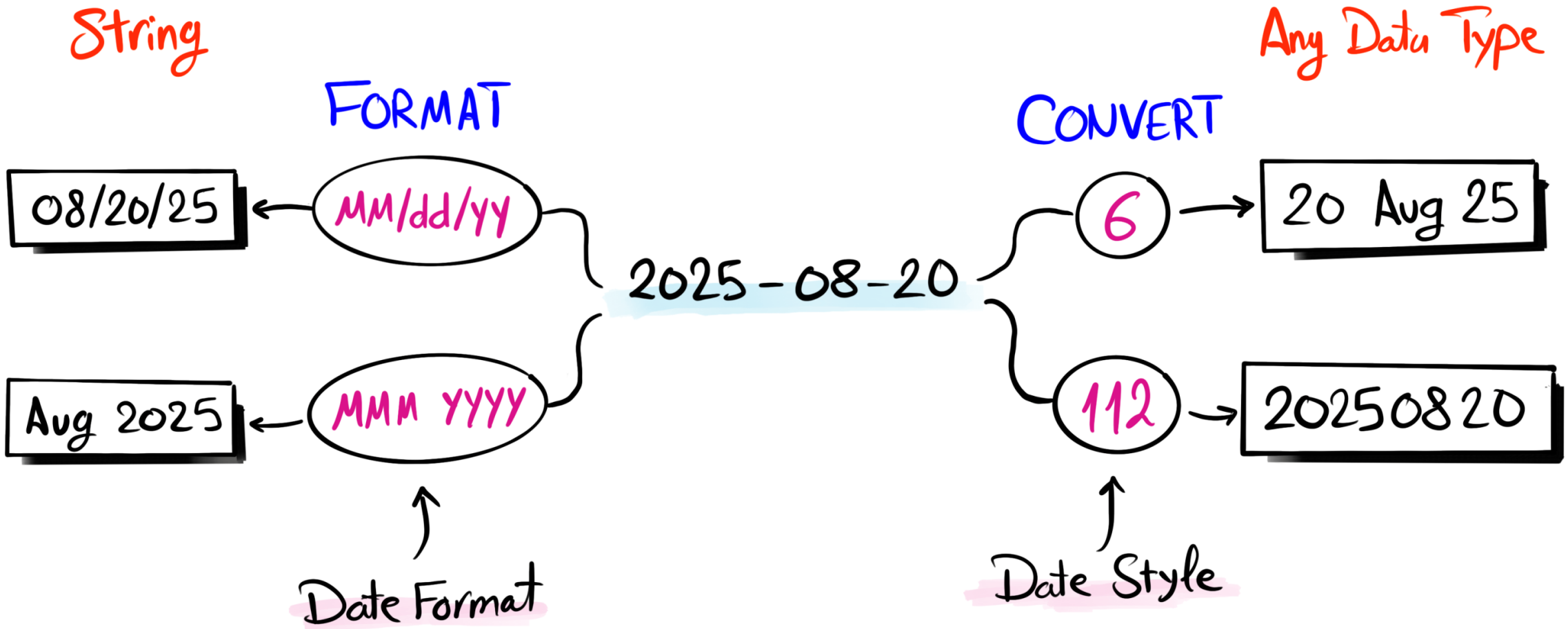
Optional

Examples

```
CONVERT(INT, '124')
```

```
CONVERT(VARCHAR, OrderDate, '34')
```

Default Style = 0


String
FORMAT
08/20/25
MM/dd/yy
Aug 2025
MMM yyyy
Date Format
2025-08-20
CONVERT
Any Data Type
6
20 Aug 25
112
20250820
Date Style

Date

#	Format	Example
1	mm/dd/yy	12/30/25
2	yy.mm.dd	25.12.30
3	dd/mm/yy	30/12/2025
4	dd.mm.yy	30.12.25
5	dd-mm-yy	30/12/2025
6	dd-Mon-yy	30-Dec-25
7	Mon dd, yy	Dec 30, 25
10	mm-dd-yy	12-30-25
11	yy/mm/dd	25/12/1930
12	yymmdd	251230
23	yyyy-mm-dd	30/12/2025
31	yyyy-dd-mm	2025-30-12
32	mm-dd-yyyy	12-30-2025
33	mm-yyyy-dd	12-2025-30
34	dd-mm-yyyy	30/12/2025
35	dd-yyyy-mm	30-2025-12
101	mm/dd/yyyy	12/30/2025
102	yyyy.mm.dd	2025.12.30
103	dd/mm/yyyy	30/12/2025
104	dd.mm.yyyy	30.12.2025
105	dd-mm-yyyy	30/12/2025
106	dd Mon yyyy	30-Dec-25
107	Mon dd, yyyy	Dec 30, 2025
110	mm-dd-yyyy	12-30-2025
111	yyyy/mm/dd	30/12/2025
112	yyyymmdd	20251230

Time

#	Format	Example
8	hh:mm:ss	00:38:54
14	hh:mm:ss:nnn	00:38:54:840
24	hh:mm:ss	00:38:54
108	hh:mm:ss	00:38:54
114	hh:mm:ss:nnn	00:38:54:840

Date & Time Styles

CONVERT

Datetime2

DATA WITH BARAA

#	Format	Example
0	Mon dd yyyy hh:mm AM/PM	Dec 30 2025 12:38AM
9	Mon dd yyyy hh:mm:ss:nnn AM/PM	Dec 30 2025 12:38:54:840AM
13	dd Mon yyyy hh:mm:ss:nnn AM/PM	30 Dec 2025 00:38:54:840AM
20	yyyy-mm-dd hh:mm:ss	2025-12-30 00:38:54
21	yyyy-mm-dd hh:mm:ss:nnn	2025-12-30 00:38:54.840
22	mm/dd/yy hh:mm:ss AM/PM	12/30/25 12:38:54 AM
25	yyyy-mm-dd hh:mm:ss:nnn	2025-12-30 00:38:54.840
26	yyyy-dd-mm hh:mm:ss:nnn	2025-30-12 00:38:54.840
27	mm-dd-yyyy hh:mm:ss:nnn	12-30-2025 00:38:54.840
28	mm-yyyy-dd hh:mm:ss:nnn	12-2025-30 00:38:54.840
29	dd-mm-yyyy hh:mm:ss:nnn	30-12-2025 00:38:54.840
30	dd-yyyy-mm hh:mm:ss:nnn	30-2025-12 00:38:54.840
100	Mon dd yyyy hh:mm AM/PM	Dec 30 2025 12:38AM
109	Mon dd yyyy hh:mm:ss:nnn AM/PM	Dec 30 2025 12:38:54:840AM
113	dd Mon yyyy hh:mm:ss:nnn	30 Dec 2025 00:38:54:840
120	yyyy-mm-dd hh:mm:ss	2025-12-30 00:38:54
121	yyyy-mm-dd hh:mm:ss:nnn	2025-12-30 00:38:54.840
126	yyyy-mm-dd T hh:mm:ss:nnn	2025-12-30T00:38:54.840
127	yyyy-mm-dd T hh:mm:ss:nnn	2025-12-30T00:38:54.840

```
2025-08-20
18:55:45.840
```

CAST

Syntax

```
CAST(value AS data_type)
```

Examples

```
CAST('123' AS INT)
```

```
CAST('2025-08-20' AS DATE)
```

No format can be specified

	CASTING	FORMATING
CAST	Any Type to Any Type	X No Formating
CONVERT	Any Type to Any Type	Formates only Date & Time
FORMAT	Any Type to Only String	Formates < Date & Time, Numbers



DATA WITH BARAA

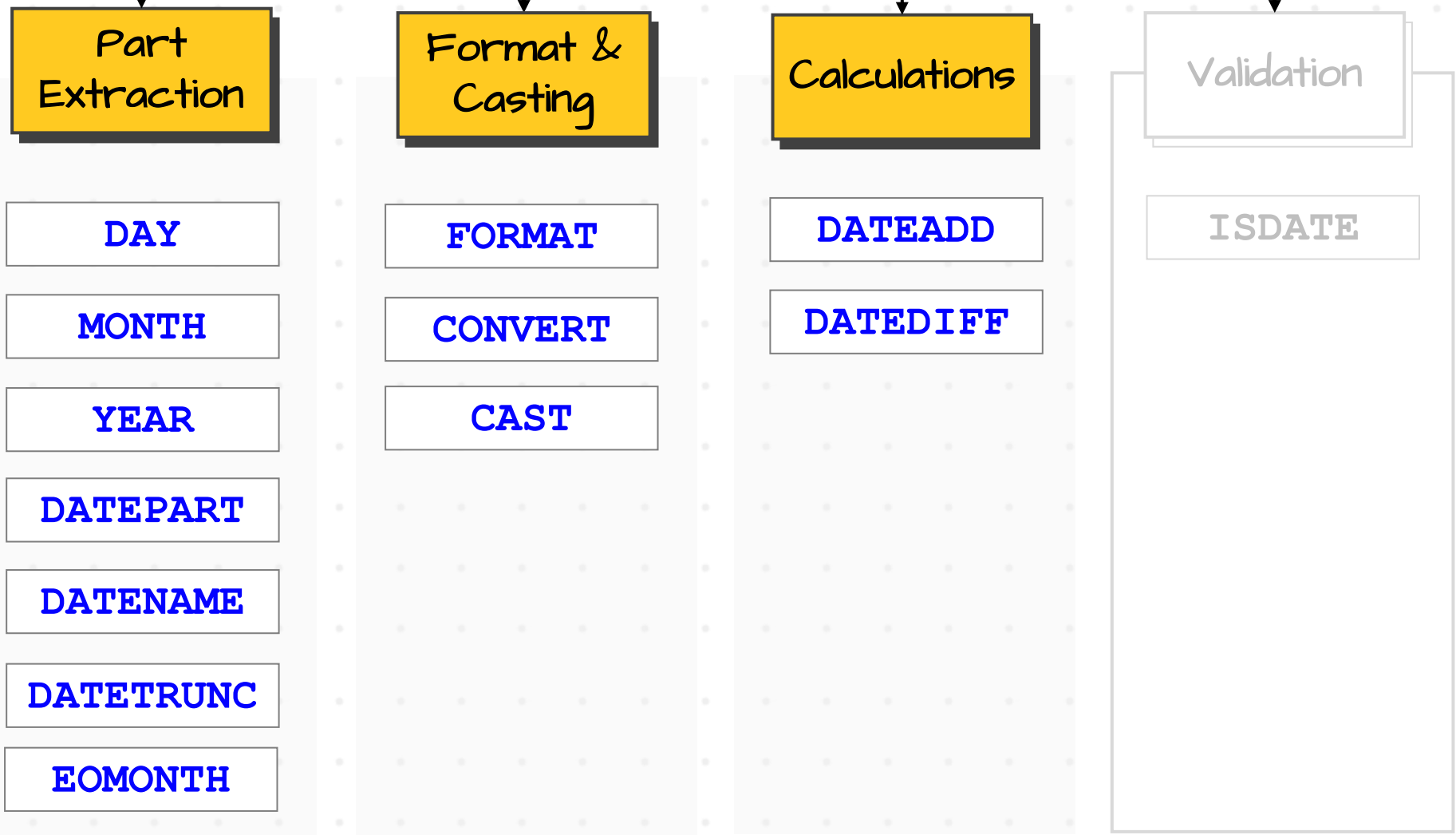
DATE CALCULATIONS

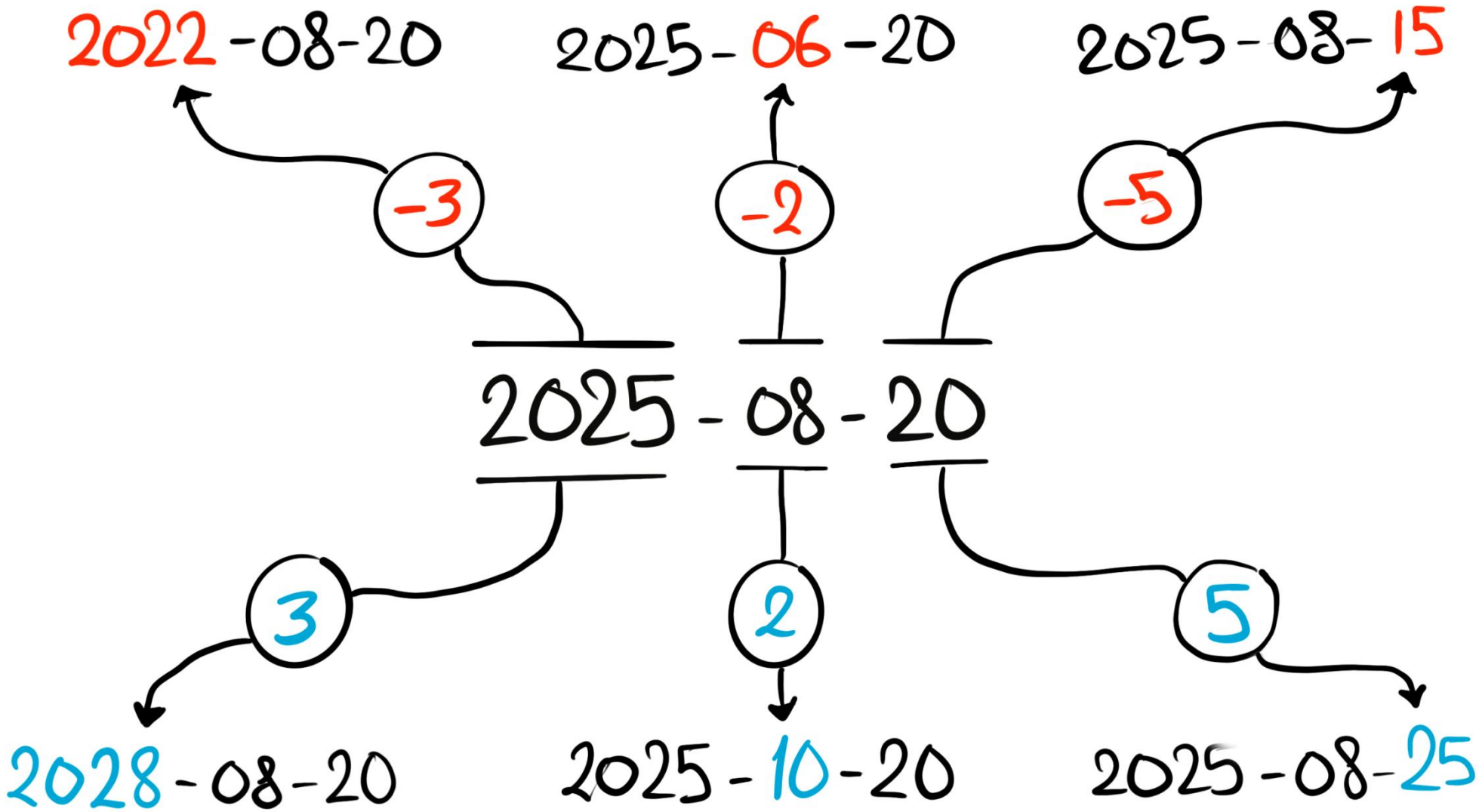
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DATA WITH BARAA
SQL Course | Date & Time Funcitons



Date & Time Functions

DATA WITH BARAA




2022-08-20
2025-06-20
2025-08-15
-3
-2
-5
2025-08-20
3
2
5
2028-08-20
2025-10-20
2025-08-25

DATEADD

Syntax

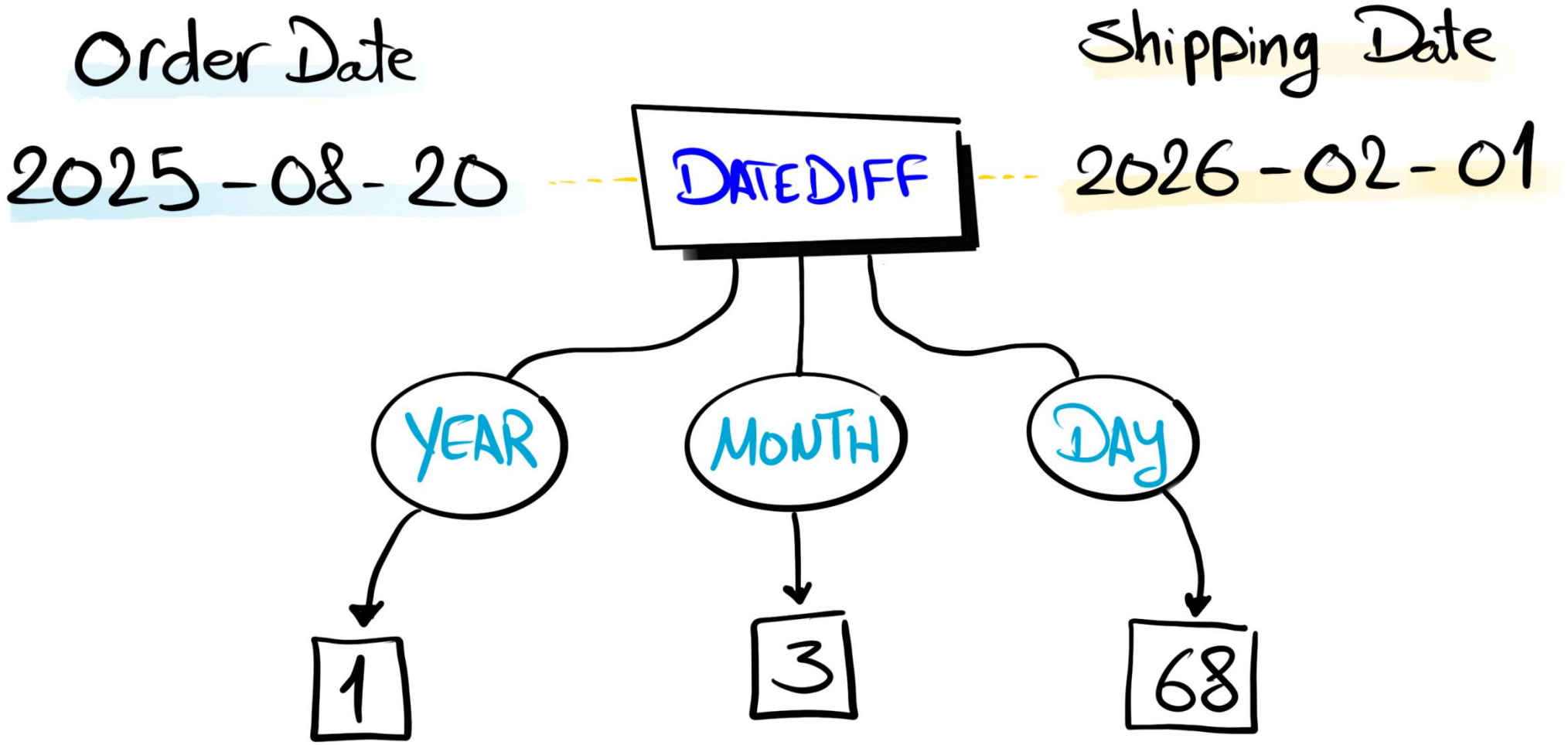
```
DATEADD(part, interval, date)
```

Examples

```
DATEADD(year, 2, OrderDate)
```

```
DATEADD(month, -4, OrderDate)
```

DATEDIFF


Order Date
2025-08-20
DATEDIFF
Shipping Date
2026-02-01
YEAR
MONTH
DAY
1
3
68

DATEDIFF

Syntax

```
DATEDIFF(part, start_date, end_date)
```

Examples

```
DATEDIFF(year, OrderDate, ShipDate)
```

```
DATEDIFF(day, OrderDate, ShipDate)
```

ISDATE

Check if a value is a date
Returns 1 if the string value is a valid date

Syntax

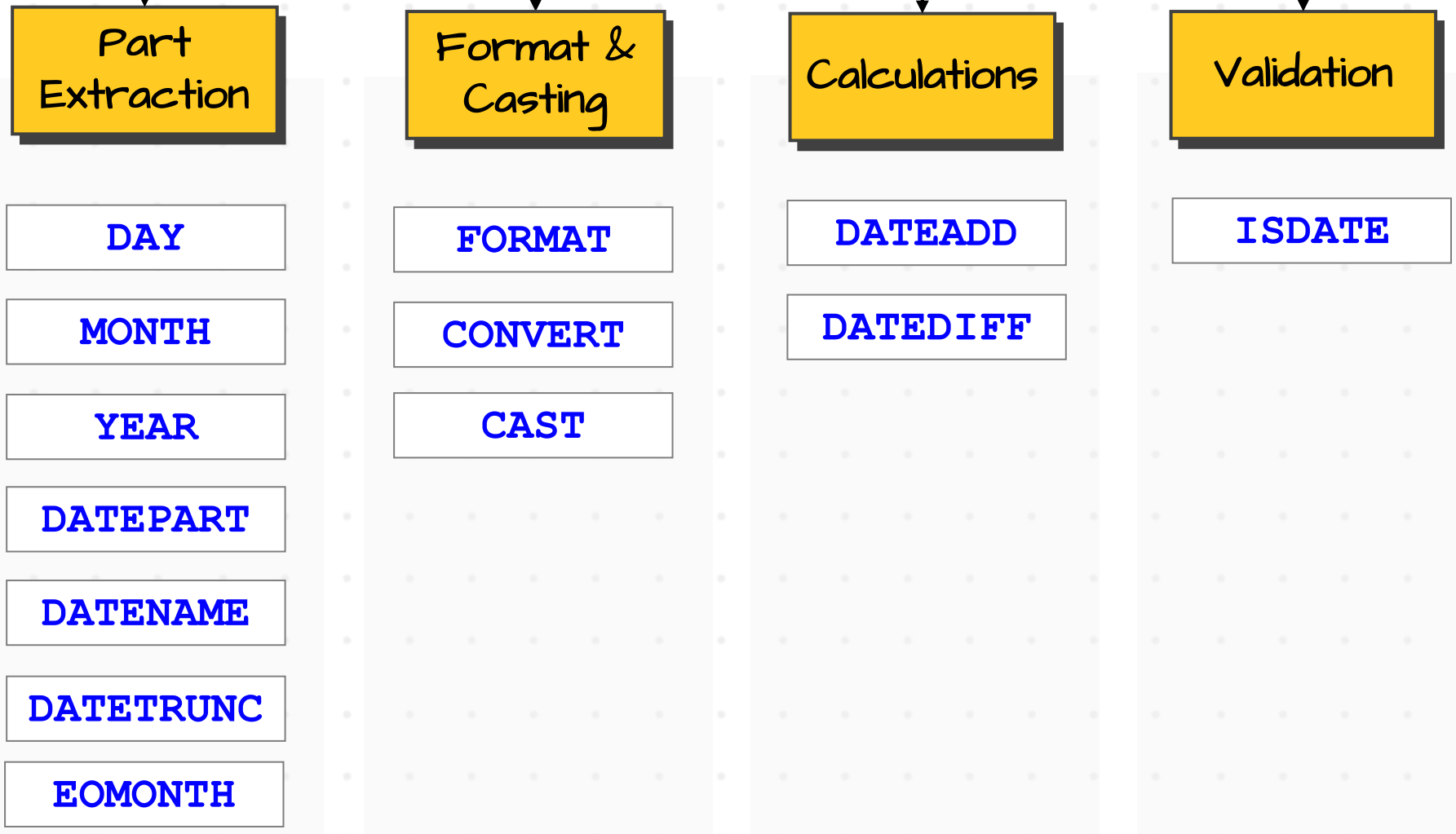
```
ISDATE(value)
```

```
ISDATE('2025-08-20')
```

```
ISDATE(2025)
```

Date & Time Functions

DATA WITH BARAA



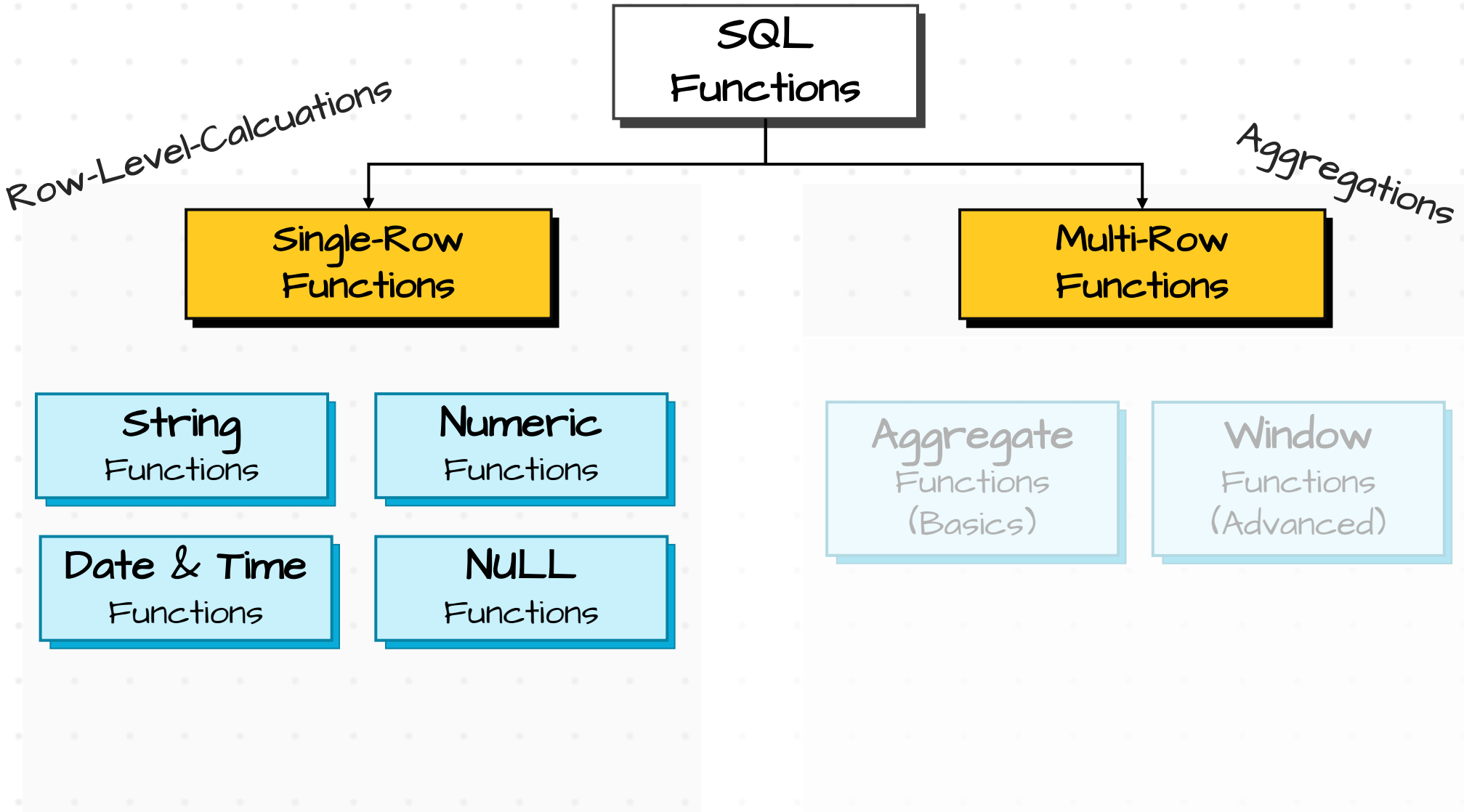


DATA WITH BARAA

NULL FUNCTIONS

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SQL Course | NULL Funcitons




SQL Functions
Row-Level-Calcuations
Aggregations
Single-Row Functions
Multi-Row Functions
String Functions
Numeric Functions
Date & Time Functions
NULL Functions
Aggregate Functions (Basics)
Window Functions (Advanced)

What are NULLS?

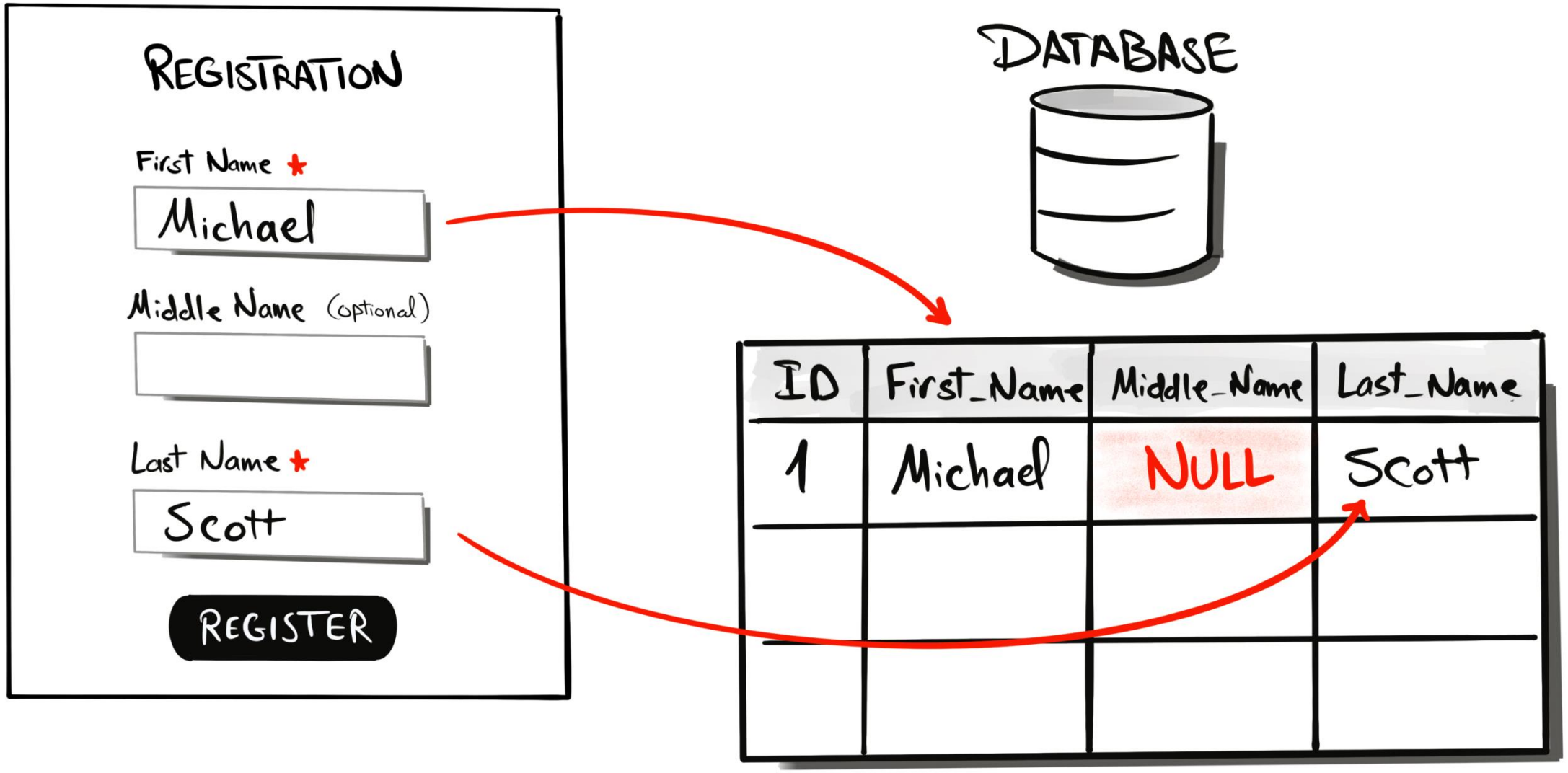
DATA WITH BARAA

ID	Name	Country	Score
1	Maria	NULL	300
2	NULL	DE	NULL
3	John	NULL	800



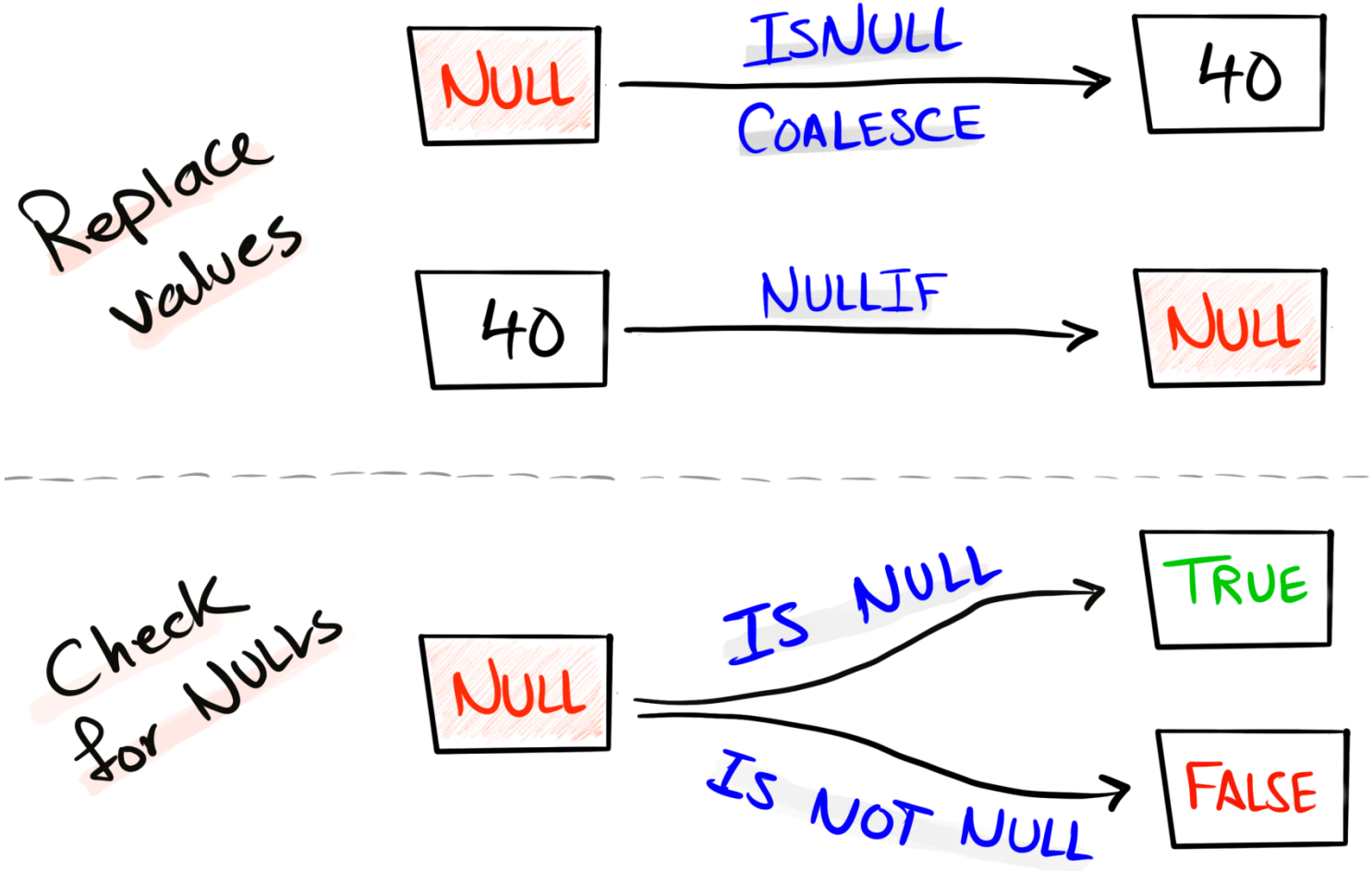
Where NULLS Come From?

DATA WITH BARAA



NULL FUNCTIONS

DATA WITH BARAA



ISNULL	COALESCE
Limited to two values	Unlimited
Fast	Slow
SQL Server → ISNULL Oracle → NVL MySQL → IFNULL	Available in All Databases

ISNULL

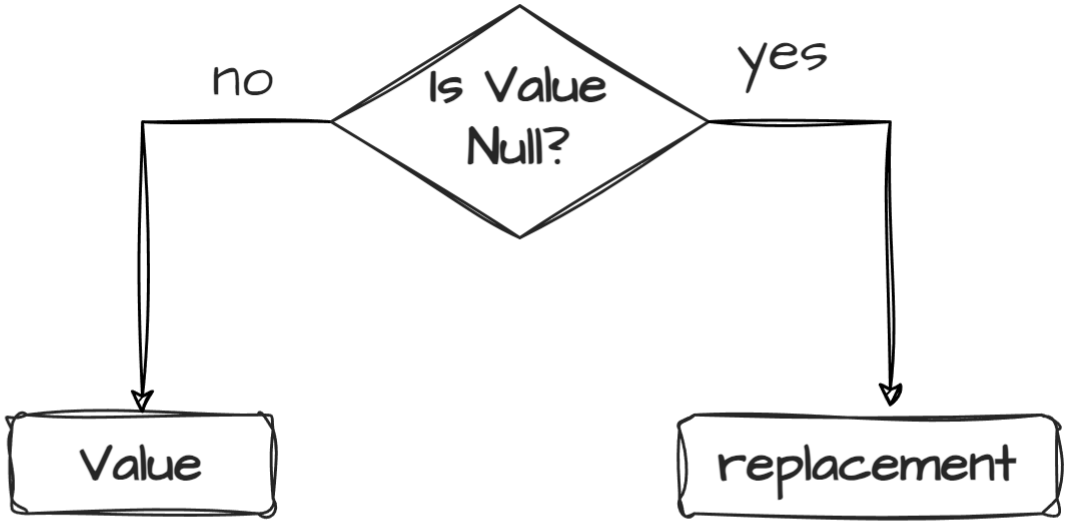
replaces NULL with the specified replacement value.

SYNTAX

```
ISNULL(value, replacement)
```

```
ISNULL(ShippingAddress, 'N/A')
```

OrderID	Shipment Address	ISNULL
1	A	A
2	NULL	N/A



ISNULL

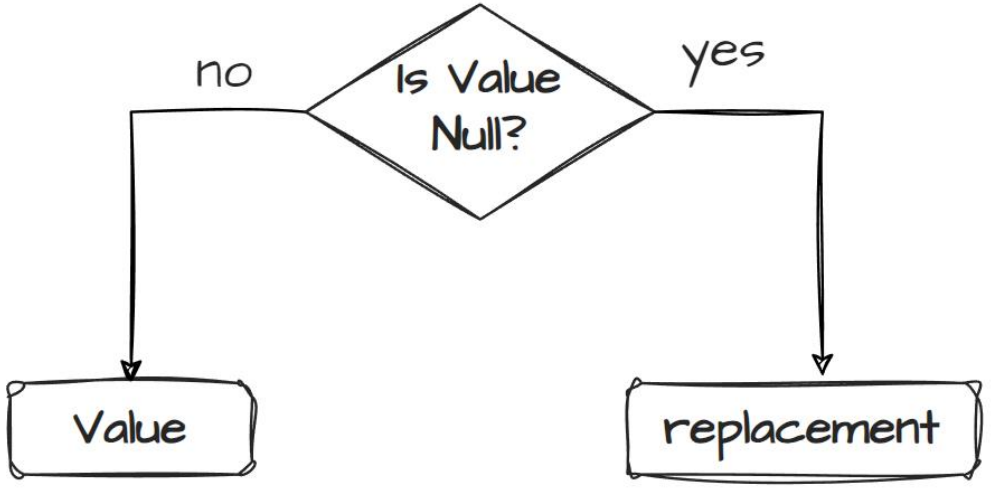
replaces NULL with the specified replacement value.

SYNTAX

```
ISNULL(value, replacement)
```

```
ISNULL(ShippingAddress, BillingAddress)
```

OrderID	Shipment Address	Billing Address	ISNULL
1	A	B	A
2	NULL	C	C
3	NULL	NULL	NULL



COALESCE

DATA WITH BARAA

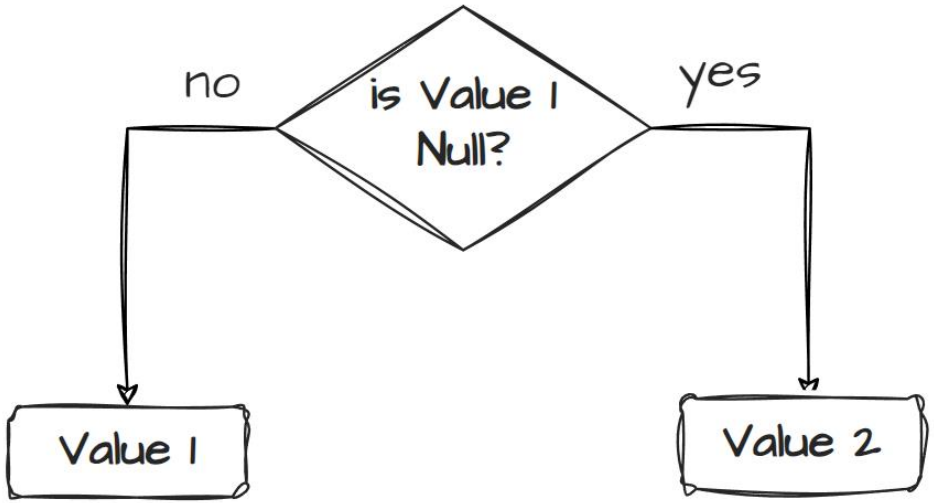
returns the first non-NULL value from the given expressions.

SYNTAX

```
COALESCE(value1, value2, value3)
```

```
COALESCE(ShippingAddress, BillingAddress)
```

OrderID	Shipment Address	Billing Address	COALESCE
1	A	B	A
2	NULL	C	C
3	NULL	NULL	NULL



COALESCE

DATA WITH BARAA

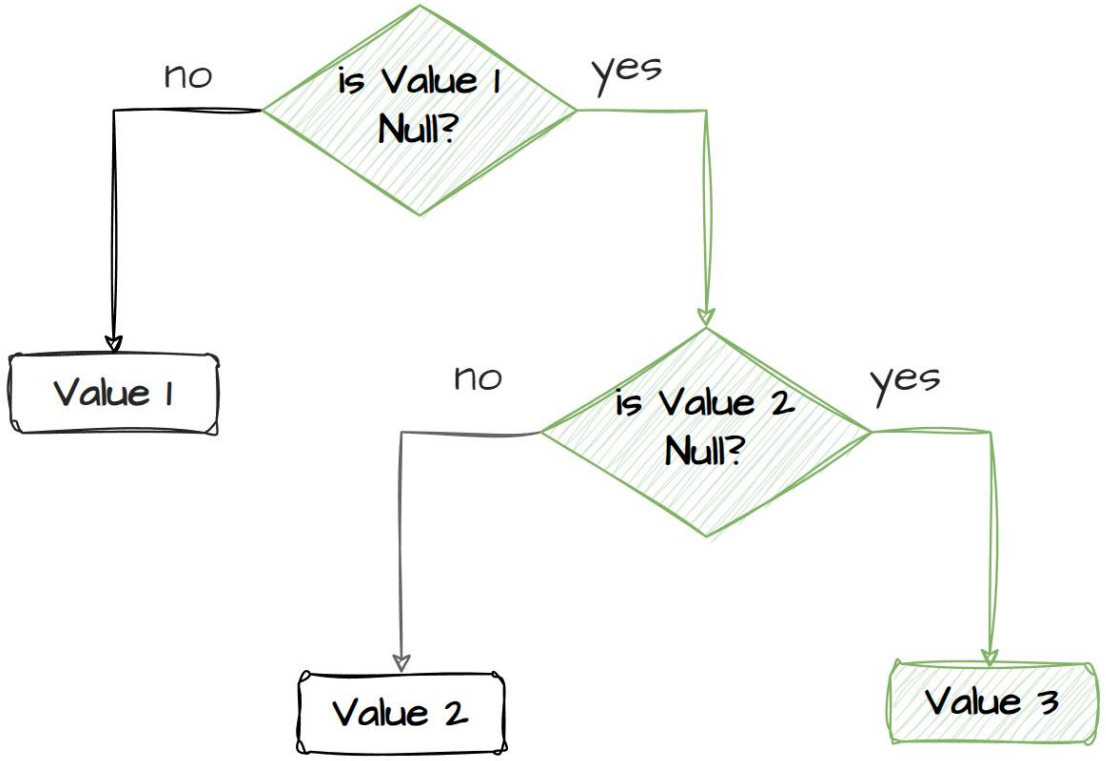
returns the first non-NULL value from the given expressions.

SYNTAX

```
COALESCE(value1, value2, value3)
```

```
COALESCE(ShippingAddress, BillingAddress, 'N/A')
```

OrderID	Shipment Address	Billing Address	COALESCE
1	A	B	A
2	NULL	C	C
3	NULL	NULL	N/A



NULLIF

DATA WITH BARAA

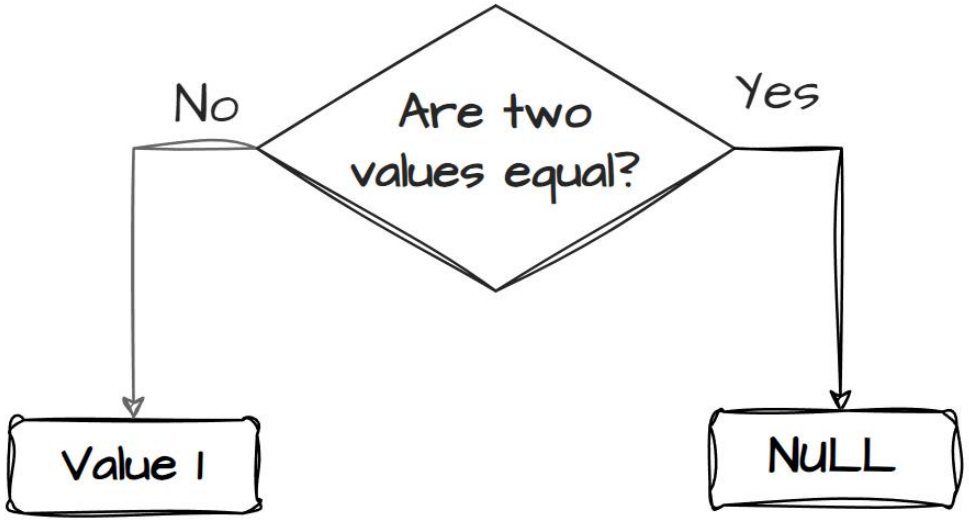
returns NULL if both values are equal; otherwise, it returns first value.

SYNTAX

```
NULLIF(value1, value2)
```

```
NULLIF(Original_Price, Discount_Price)
```

OrderID	Original_Price	Discount_Price	NULLIF
1	150	50	150
2	250	250	NULL



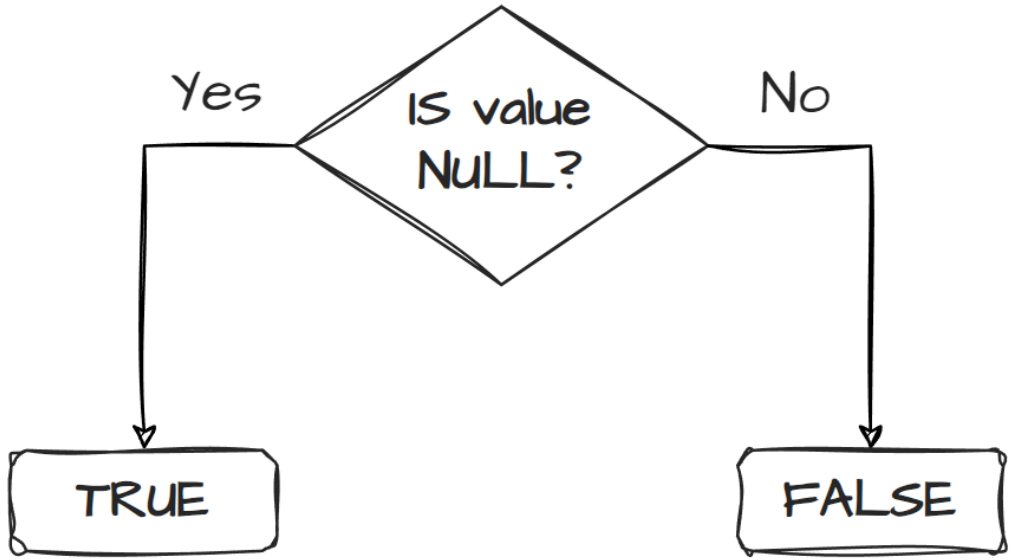
IS NULL

DATA WITH BARAA

check if a value is NULL.

```
Price IS NULL
```

OrderID	Price	IS NULL	IS NOT NULL
1	90	FALSE	TRUE
2	NULL	TRUE	FALSE



IS NULL

DATA WITH BARAA

In SQL, use IS NULL instead of = NULL to correctly filter rows with NULL values.

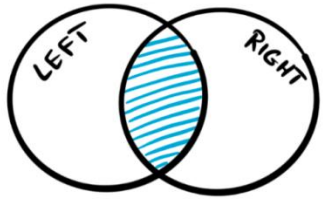


DATA WITH BARAA

JOINS & IS NULL

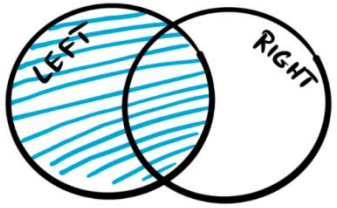
Inner Join

Only matching Rows

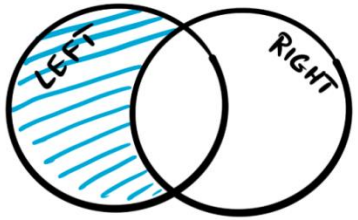


Left Join

All Rows from Left and matching rows



Left Anti Join

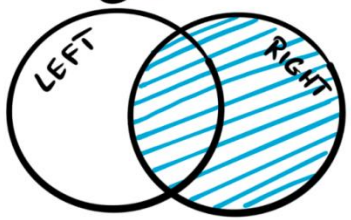


All Rows from Left without matching rows

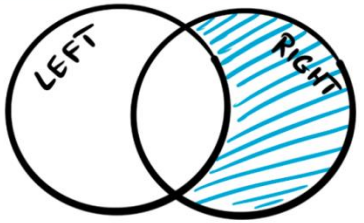
(Left Join + Is NULL)

Right Join

All Rows from right and matching rows



Right Anti Join

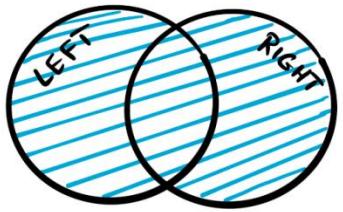


All Rows from Right without matching rows

(Right Join + Is NULL)

Full Join

All Rows



NULL vs Empty vs Blank

DATA WITH BARAA

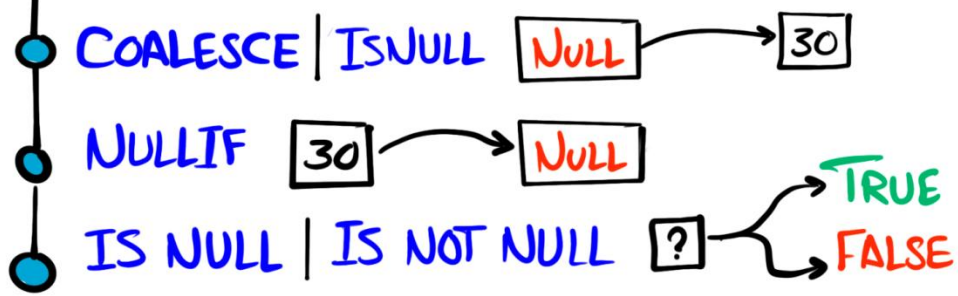
	NULL	Empty String	Blank Space
Representation	NULL	''	' '
Meaning	unknown	Known, Empty Value	Known, Space Value
Data Type	Special Marker	String (0)	String (1 or more)
Storage	Very minimal	occupies memory	occupies memory (each space)
Performance	Best	Fast	Slow
Comparison	IS NULL	= ''	= ' '

NULL Functions

DATA WITH BARAA

- Nulls special markers means missing value.
- Using Nulls can optimize storage and performance.

Functions



USE CASES

- Handle Nulls - Data Aggregation
- Handle Nulls - Mathematical operations
- Handle Nulls - Joining Tables
- Handle Nulls - Sorting Data
- Finding unmatched data - Left Anti Join
- Data Policies → Nulls, Default Value



DATA WITH BARAA

CASE STATEMENT

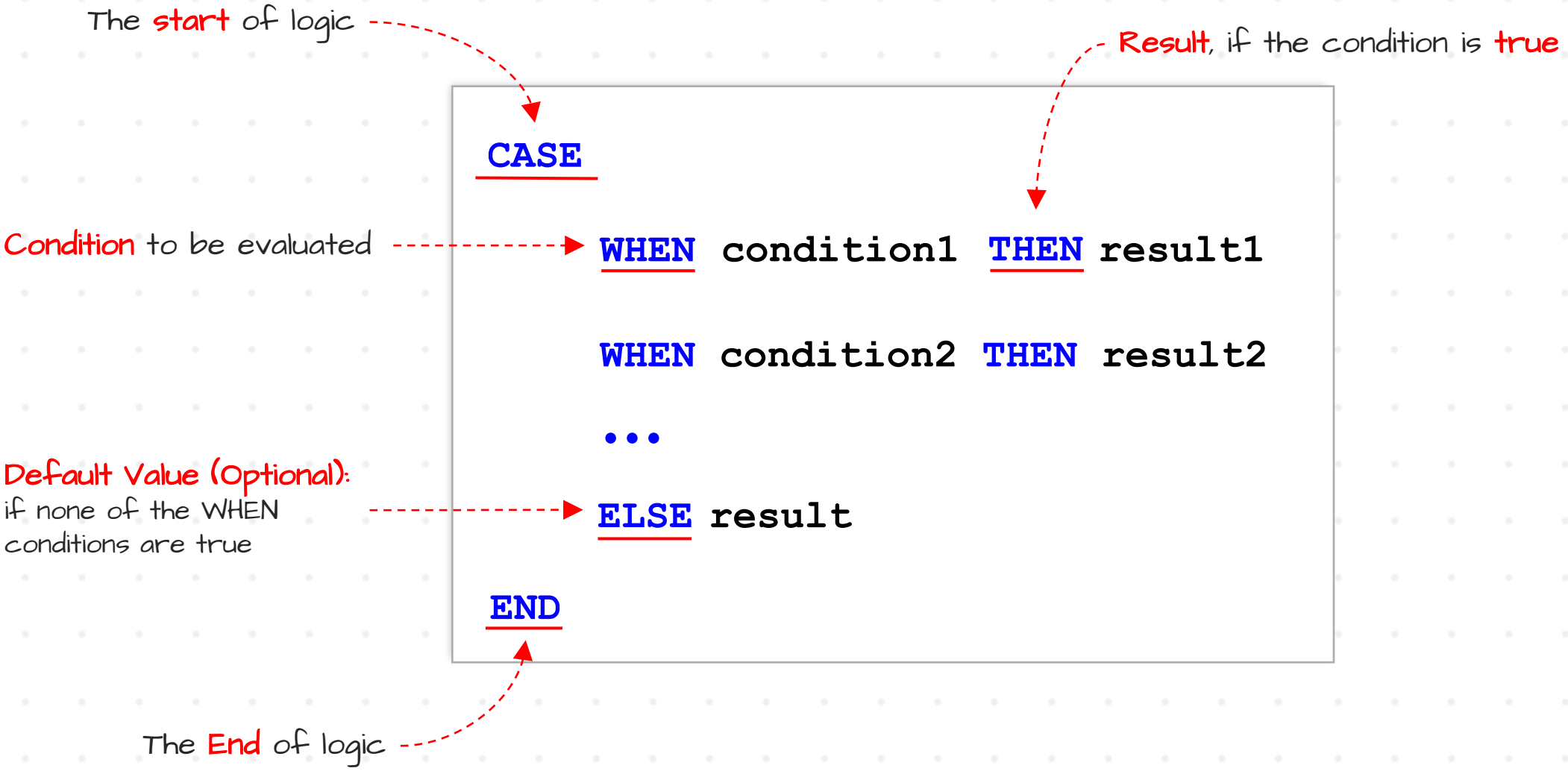
CASE WHEN

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SQL Course | Case Statement



CASE STATEMENT
Syntax

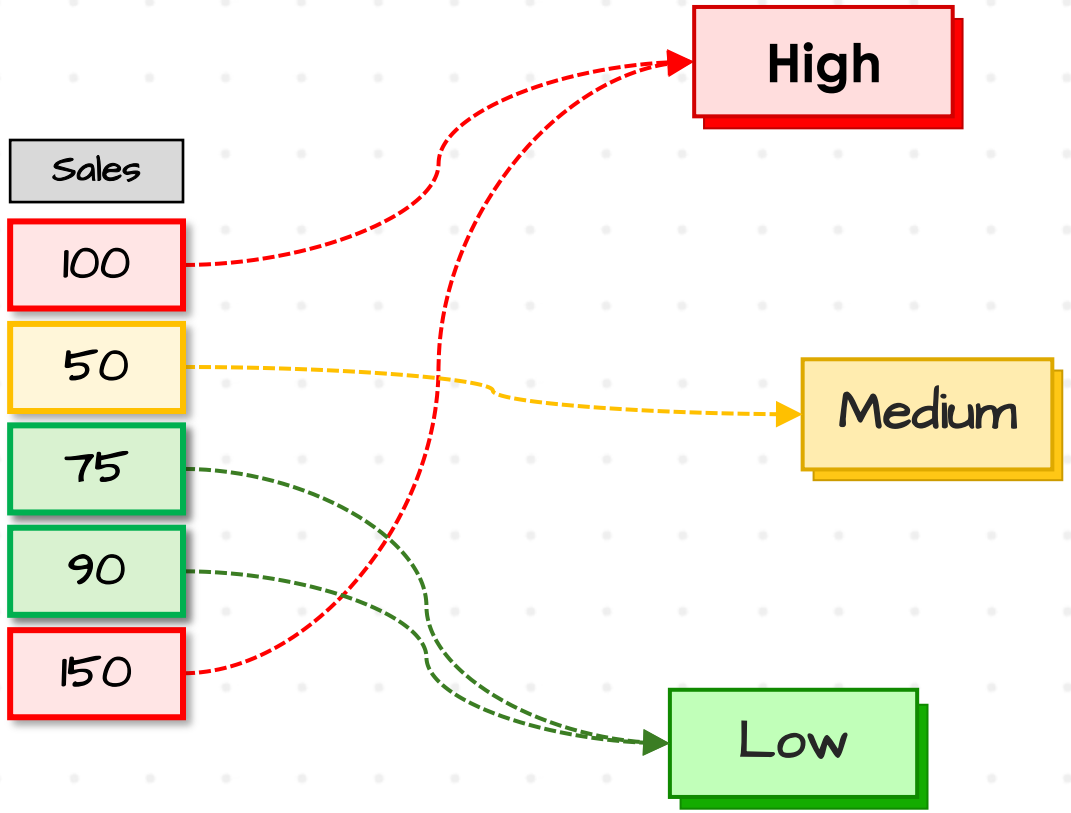
DATA WITH BARAA



Use Case: Derive New Columns

The CASE statement in SQL categorizes values based on conditions

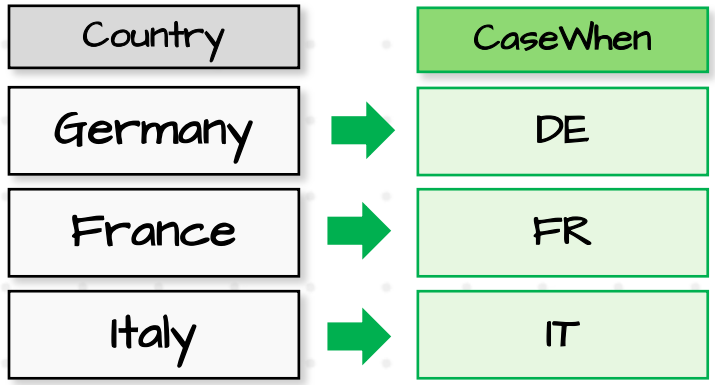
```
CASE
     WHEN Sales >= 100 THEN 'High'
     WHEN Sales >= 50 THEN 'Medium'
     ELSE 'Low'
END
```



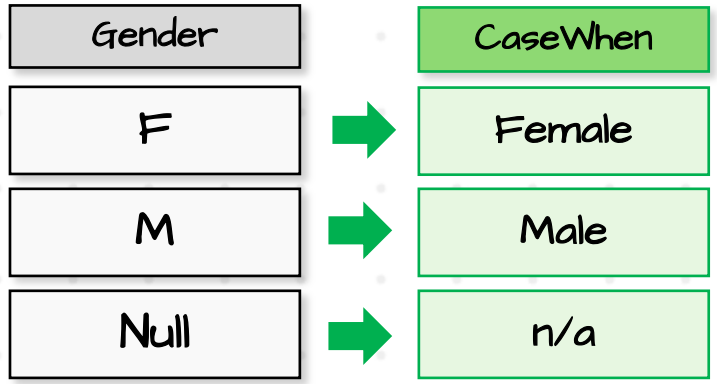
Transformation & Standardization

The CASE statement in SQL is used for data transformation and standardization by mapping specific values to standardized formats.

```sql
CASE
    WHEN Country = 'Germany' THEN 'DE'
    WHEN Country = 'France'  THEN 'FR'
    WHEN Country = 'Italy'   THEN 'IT'
ELSE 'n/a'
END
```




```sql
CASE
    WHEN Country = 'F'  THEN 'Female'
    WHEN Country = 'M'  THEN 'Male'
ELSE 'n/a'
END
```



Column Name
to be evaluated (Only One)

Column Value
To be compared

```sql
CASE
    WHEN Country = 'Germany' THEN 'DE'
    WHEN Country = 'India'   THEN 'IN'
    WHEN Country = 'United States' THEN 'US'
    WHEN Country = 'France'  THEN 'FR'
    WHEN Country = 'Italy'   THEN 'IT'
    ELSE 'n/a'
END
```

Full Form

```sql
CASE Country
    WHEN 'Germany' THEN 'DE'
    WHEN 'India'   THEN 'IN'
    WHEN 'United States' THEN 'US'
    WHEN 'France'  THEN 'FR'
    WHEN 'Italy'   THEN 'IT'
    ELSE 'n/a'
END
```

Quick Form

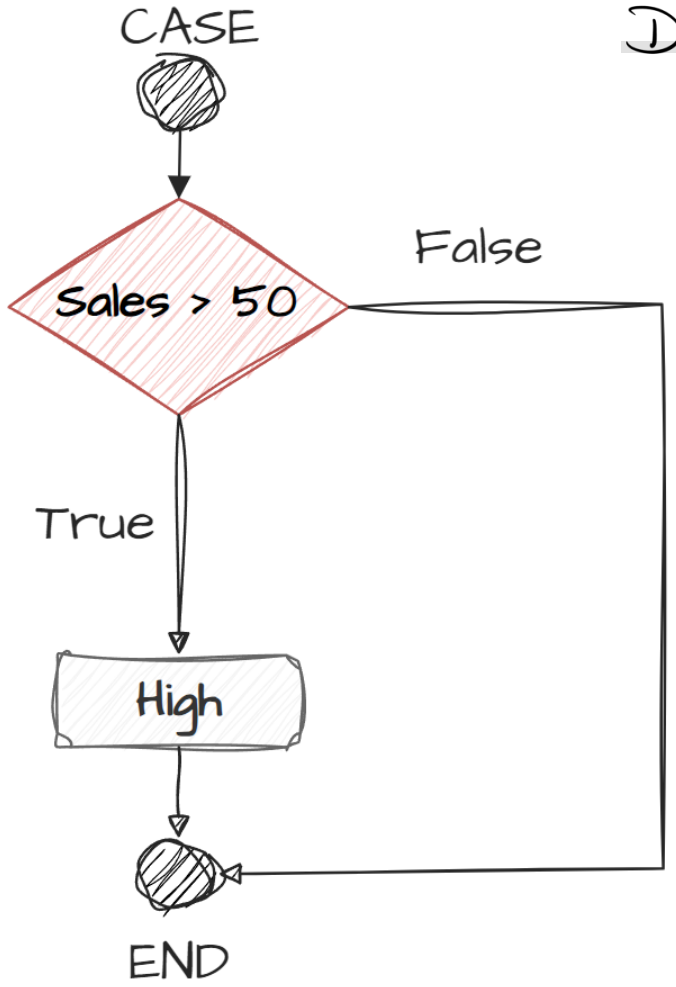
DATA WITH BARAA

```
CASE

  WHEN Sales > 50 THEN 'High'

END
```

Sales	CASE
60	High
30	NULL
15	NULL
NULL	NULL



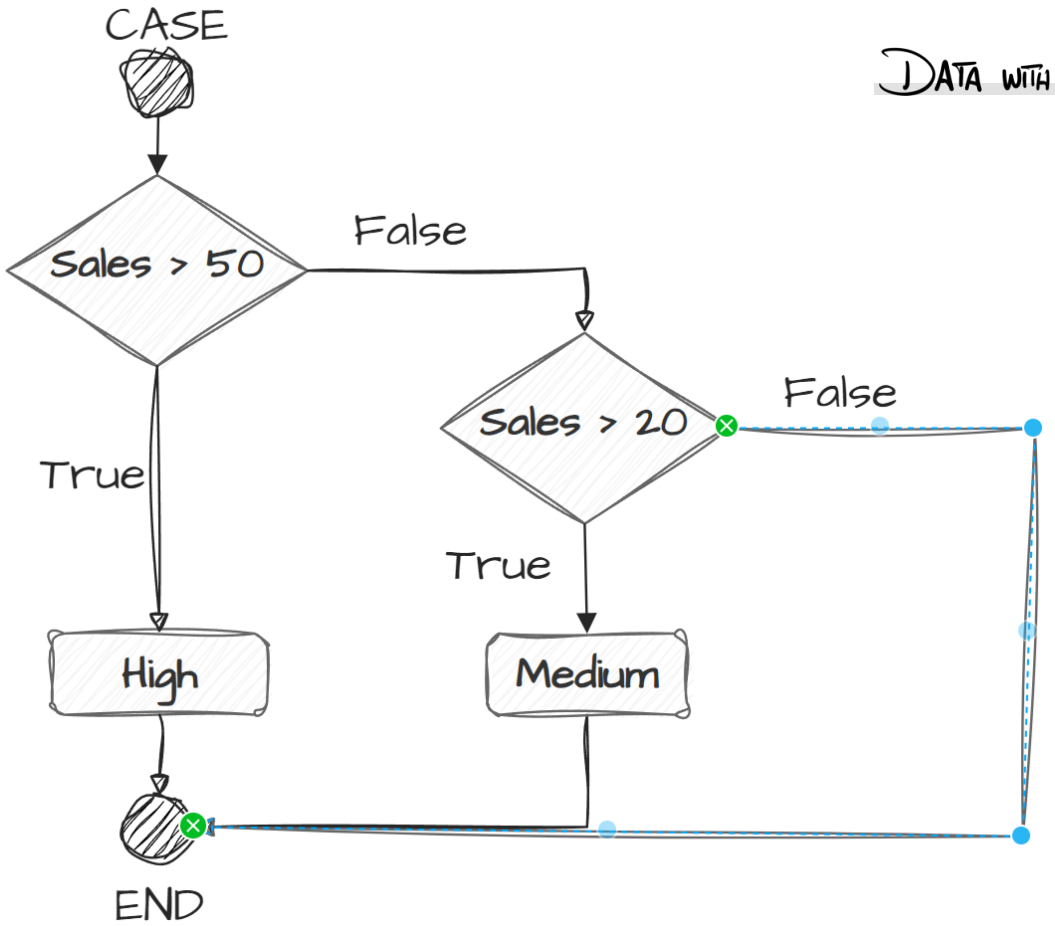
```sql
CASE

  WHEN Sales > 50 THEN 'High'

  WHEN Sales > 20 THEN 'Medium'

END
```

Sales	CASE
60	High
30	Medium
15	NULL
NULL	NULL



```sql
CASE

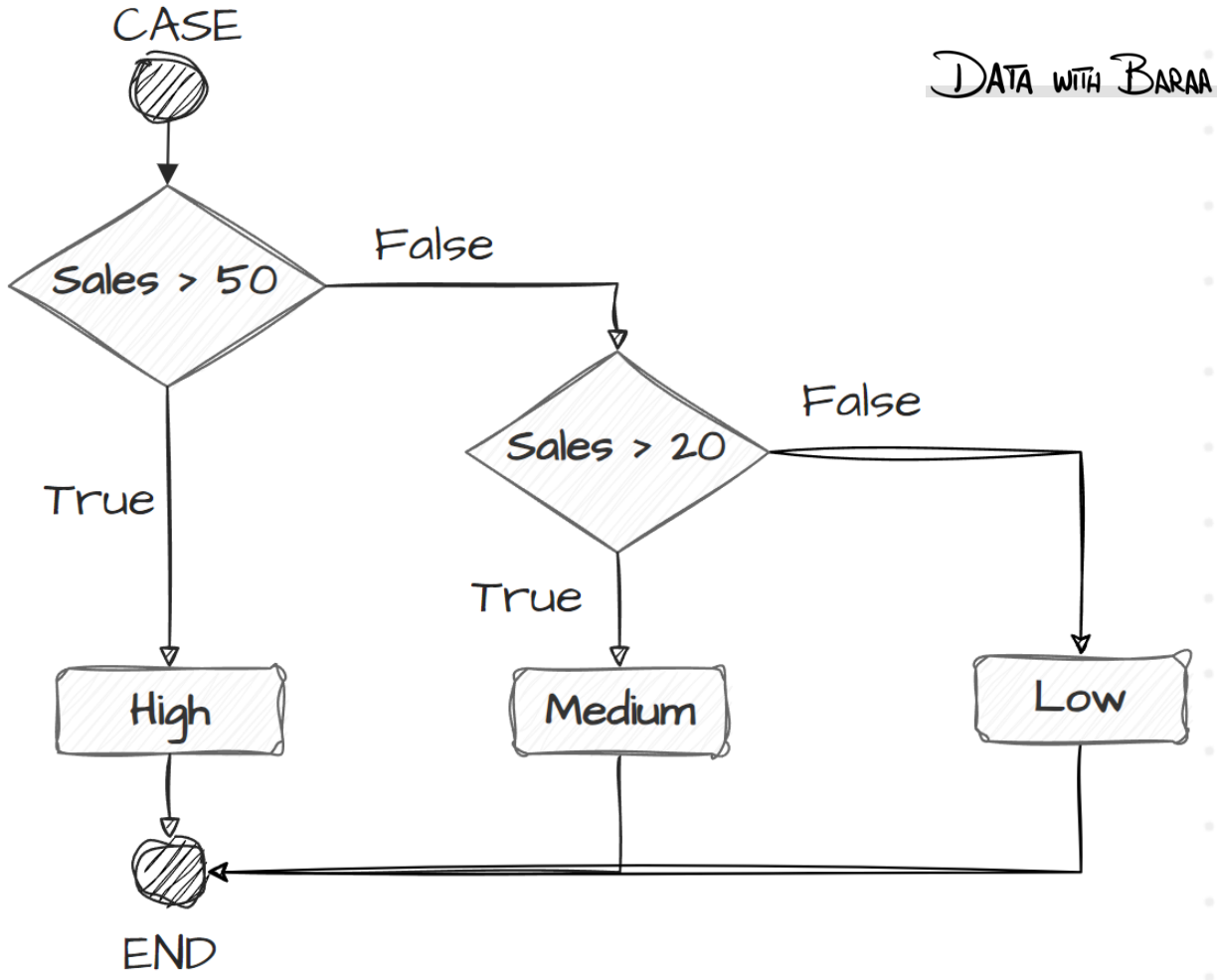
  WHEN Sales > 50 THEN 'High'

  WHEN Sales > 20 THEN 'Medium'

  ELSE 'Low'

END
```

Sales	CASE
60	High
30	Medium
15	Low
NULL	Low



CASE STATEMENT

DATA WITH BARAA

Evaluates a list of conditions and returns a value when the first Condition is met.

Rules The data type of the results must be matching

USE CASES

- Derive New Information
  - Categorizing Data
  - Mapping Values
  - Handling NULLs
  - Conditional Aggregations